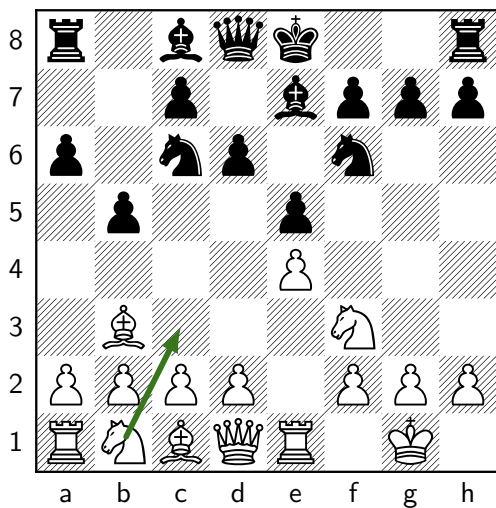


Melanie-ANALOGY — Ruy Lopez, Closed

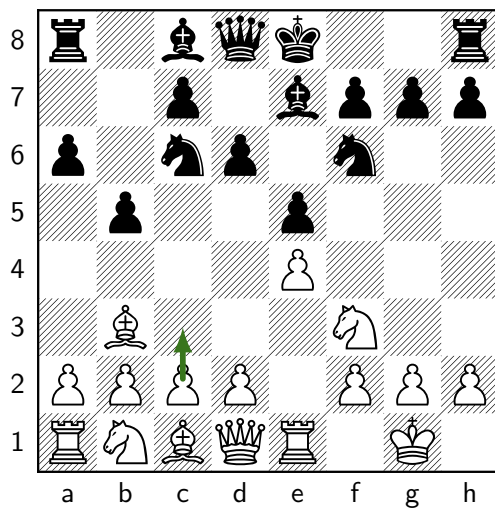
Melanie (Deep_Vision)

Abstract

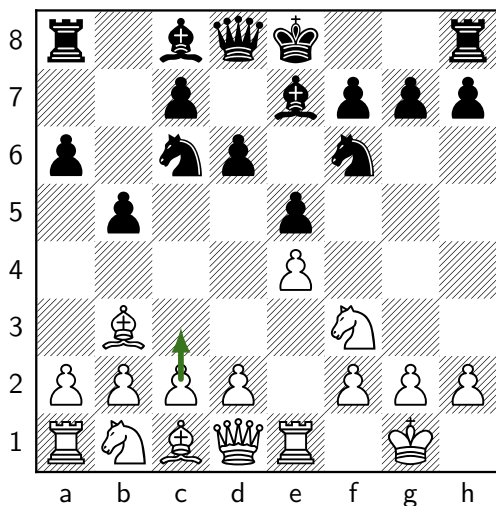
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



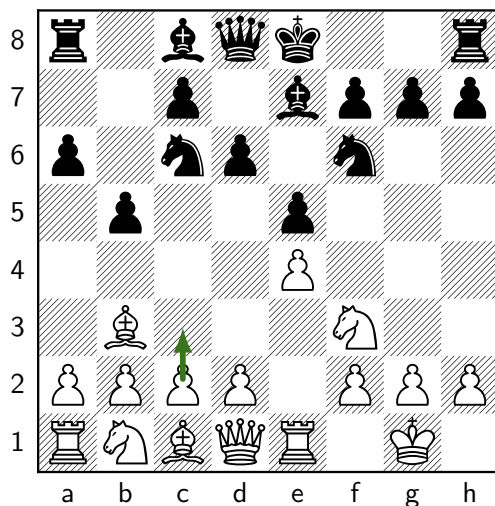
(a)



(b)

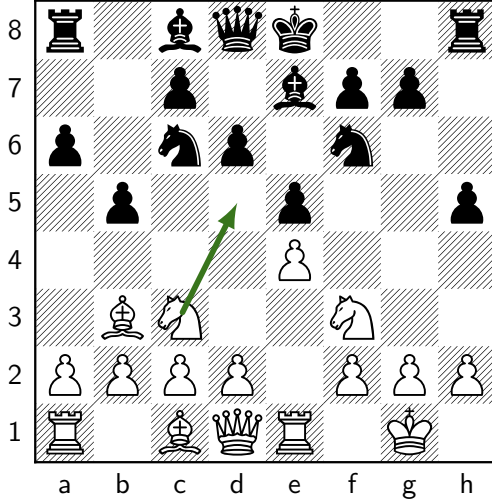


(c)

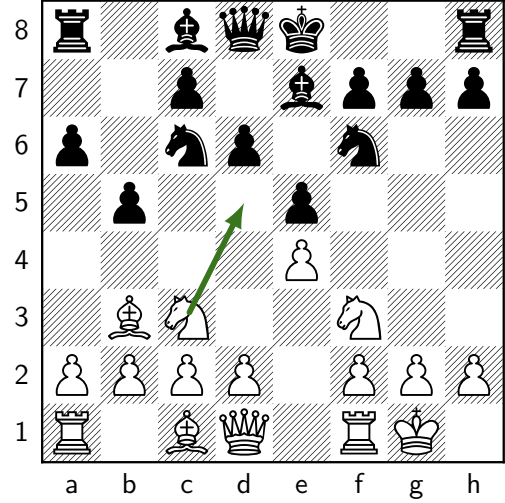


(d)

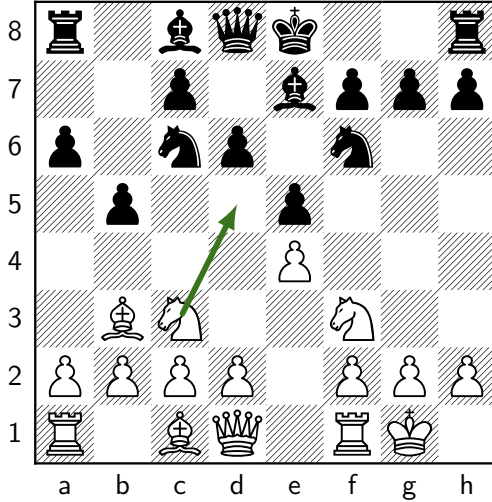
Figure 1: **Move 8, White: Nc3**. (a) Our position; Melanie plays Nc3 (b1c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#152` (ply 14) [structural distance $d=0$, QAP obj=80]: the master played c3; under μ our Nb1 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Nb1; transferred move b1c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#153` (ply 14) [structural distance $d=0$, QAP obj=80]: the master played c3; under μ our Nb1 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Nb1; transferred move b1c3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#168` (ply 14) [structural distance $d=0$, QAP obj=80]: the master played c3; under μ our Nb1 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Nb1; transferred move b1c3 (**endorses this move**). Synthesis: 62 retrieved analogs transferred to Nc3 (summed QAP objective 4968, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



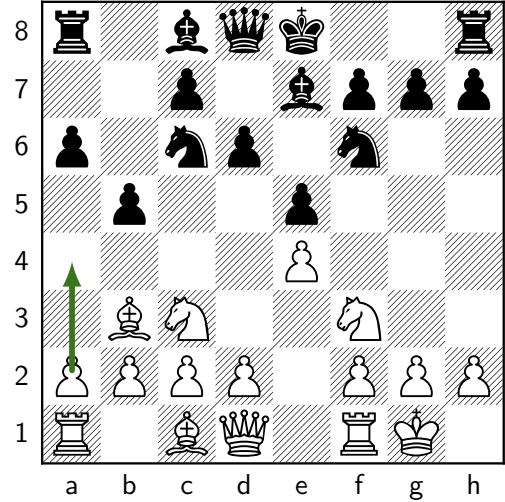
(a)



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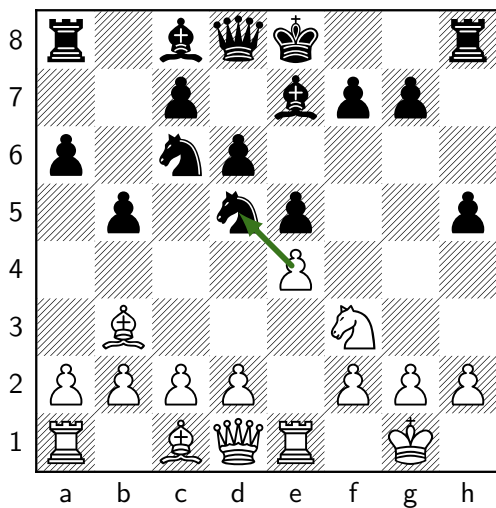


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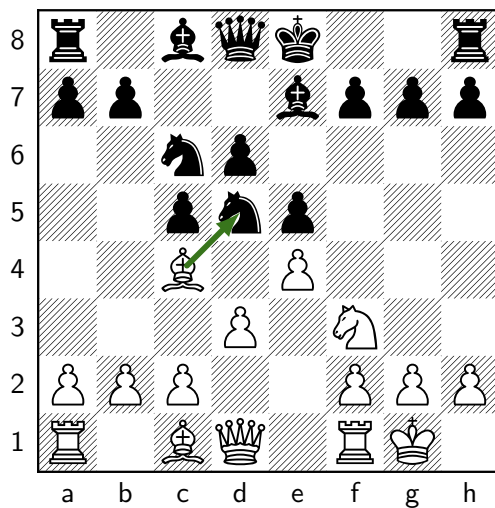


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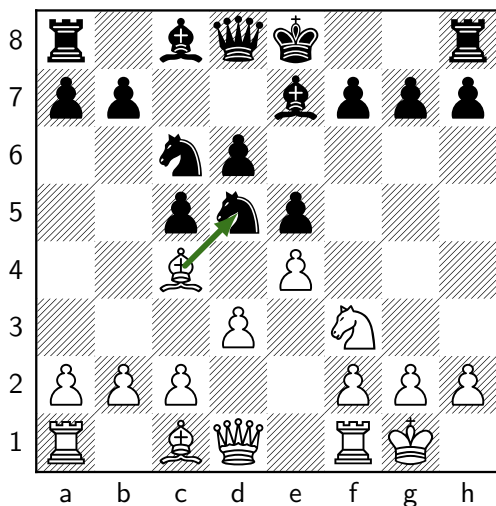
Figure 2: **Move 9, White: Nd5.** (a) Our position; Melanie plays Nd5 (c3d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3466` (ply 14) [structural distance $d=5$, QAP obj=95]: the master played Nd5; under μ our $Nc3$ plays the role of that game’s $Nc3$; cross-type roles: our $Nc3$ as their $Bb3$; transferred move c3d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3467` (ply 14) [structural distance $d=5$, QAP obj=95]: the master played Nd5; under μ our $Nc3$ plays the role of that game’s $Nc3$; cross-type roles: our $Nc3$ as their $Bb3$; transferred move c3d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2416` (ply 14) [structural distance $d=5$, QAP obj=95]: the master played a4; under μ our $Pa2$ plays the role of that game’s $Pa2$; cross-type roles: our $Nc3$ as their $Bb3$; transferred move a2a4. Synthesis: 2 retrieved analogs transferred to Nd5 (summed QAP objective 190, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



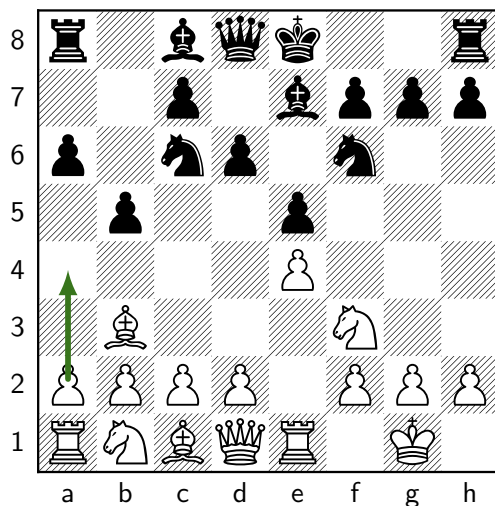
(a)



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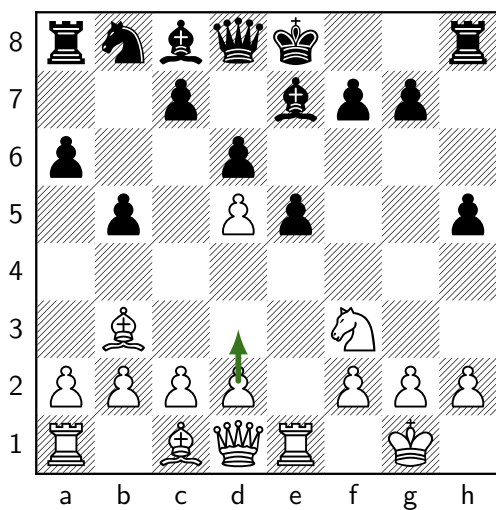


(c)

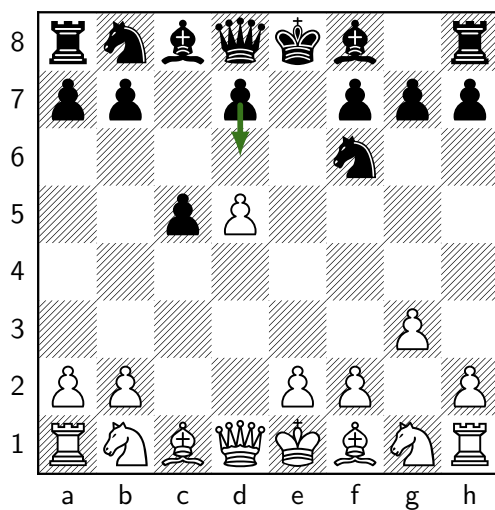


(d)

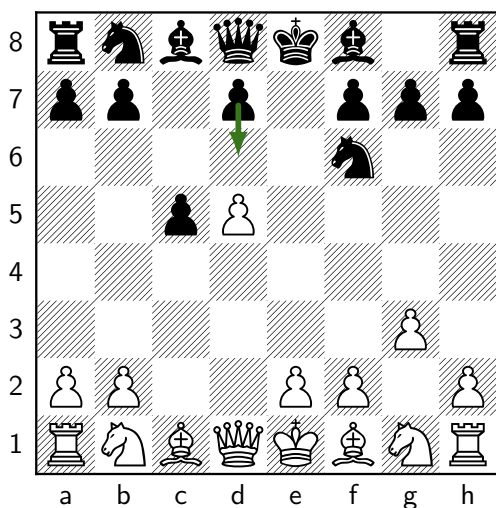
Figure 3: **Move 10, White: exd5.** (a) Our position; Melanie plays **exd5** (**e4d5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#949` (ply 14) [structural distance $d=34$, QAP obj=44]: the master played **Bxd5**; under μ our *Pe4* plays the role of that game’s *Bc4* (a **cross-type** correspondence); cross-type roles: our *Bc1* as their *Ra1*, our *Pc2* as their *Bc1*; transferred move **e4d5** (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#950` (ply 14) [structural distance $d=34$, QAP obj=44]: the master played **Bxd5**; under μ our *Pe4* plays the role of that game’s *Bc4* (a **cross-type** correspondence); cross-type roles: our *Bc1* as their *Ra1*, our *Pc2* as their *Bc1*; transferred move **e4d5** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3035` (ply 14) [structural distance $d=38$, QAP obj=6]: the master played **a4**; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *Bb3* as their *Ra1*, our *Pc2* as their *Nb1*; transferred move **a2a4**. Synthesis: 2 retrieved analogs transferred to **exd5** (summed QAP objective 88, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



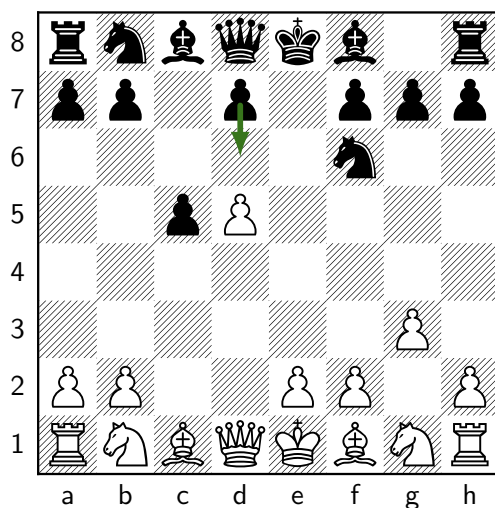
(a)



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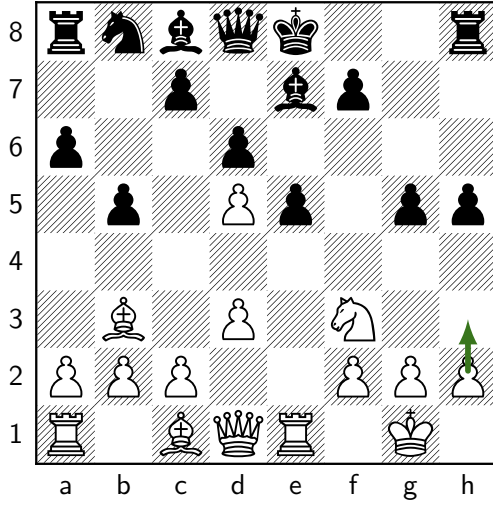


(c)

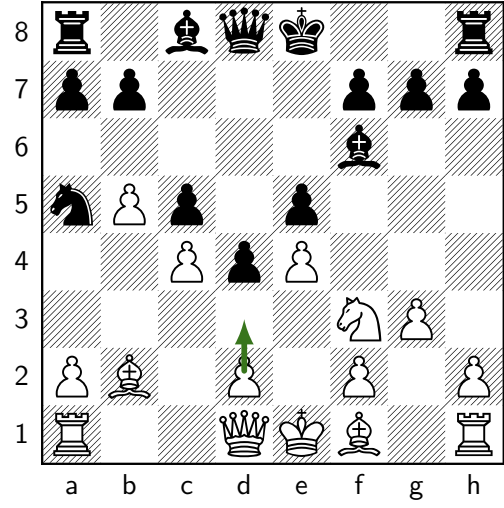


(d)

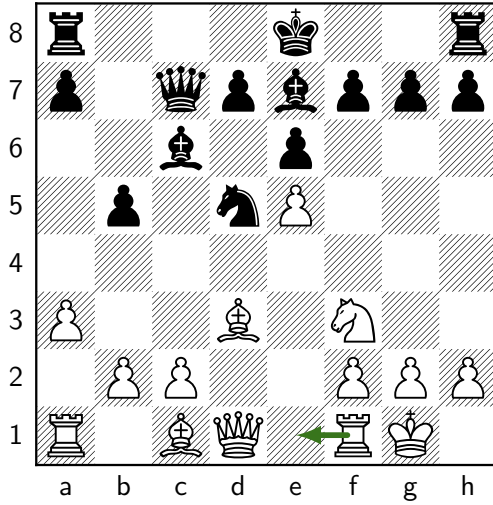
Figure 4: **Move 11, White: d3.** (a) Our position; Melanie plays d3 (d2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1220` (ply 9) [structural distance $d=61$, QAP obj=25]: the master played d6; under μ our $Pd2$ plays the role of that game’s $Pd2$; cross-type roles: our $Qd1$ as their $Ra8$, our $pb5$ as their $Nb8$; transferred move d2d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1221` (ply 9) [structural distance $d=61$, QAP obj=25]: the master played d6; under μ our $Pd2$ plays the role of that game’s $Pd2$; cross-type roles: our $Qd1$ as their $Ra8$, our $pb5$ as their $Nb8$; transferred move d2d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1574` (ply 9) [structural distance $d=61$, QAP obj=25]: the master played d6; under μ our $Pd2$ plays the role of that game’s $Pd2$; cross-type roles: our $Qd1$ as their $Ra8$, our $pb5$ as their $Nb8$; transferred move d2d3 (**endorses this move**). Synthesis: 22 retrieved analogs transferred to d3 (summed QAP objective 554, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



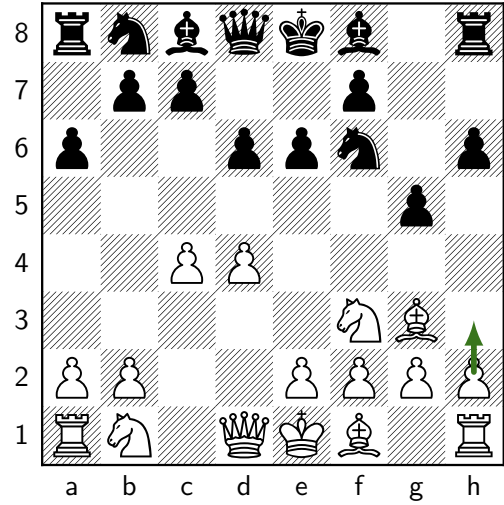
(a)



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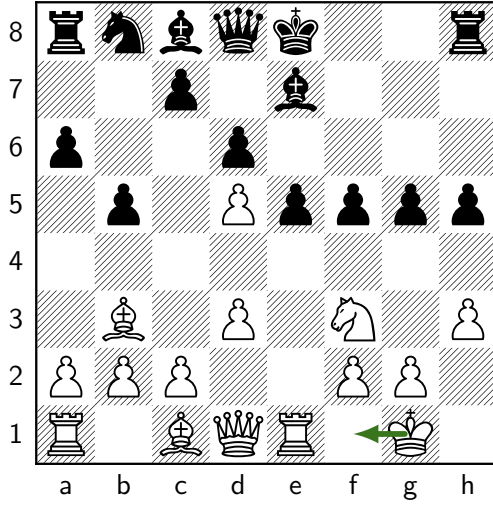


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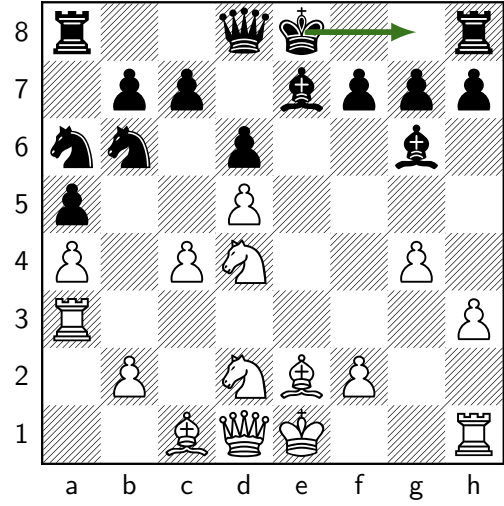


(d)

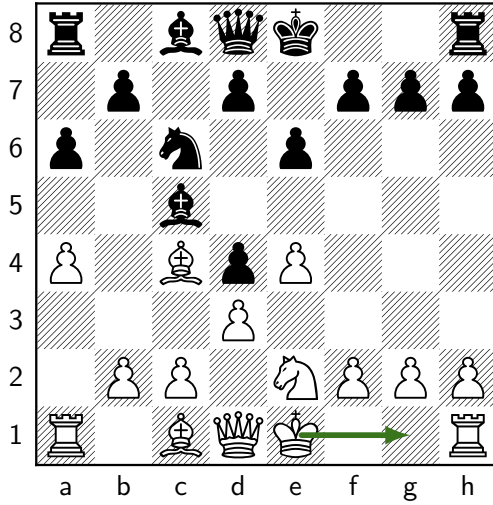
Figure 5: **Move 12, White: h3.** (a) Our position; Melanie plays **h3** (**h2h3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1986` (ply 24) [structural distance $d=70$, QAP obj=84]: the master played **d3**; under μ our *Ph2* plays the role of that game’s *Pd2*; cross-type roles: our *Bb3* as their *Qd1*, our *Pa2* as their *Rh1*; transferred move **h2h3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3795` (ply 24) [structural distance $d=72$, QAP obj=49]: the master played **Re1**; under μ our *Ph2* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *qd8* as their *Ra1*, our *Pd3* as their *Bc1*; transferred move **h2h3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#34` (ply 12) [structural distance $d=61$, QAP obj=36]: the master played **h3**; under μ our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our *Pg2* as their *Nb1*, our *pa6* as their *Qd1*; transferred move **h2h3** (**endorses this move**). Synthesis: 5 retrieved analogs transferred to **h3** (summed QAP objective 241, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



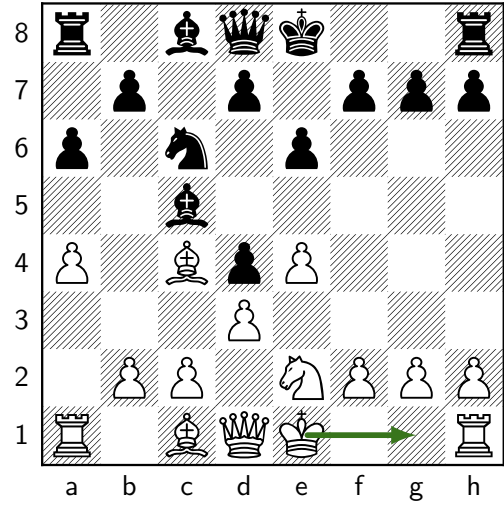
(a)



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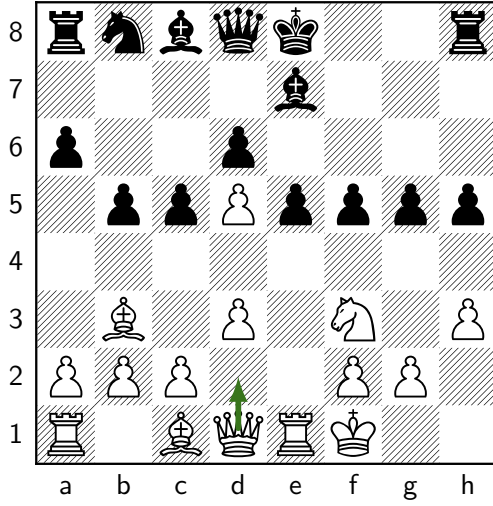


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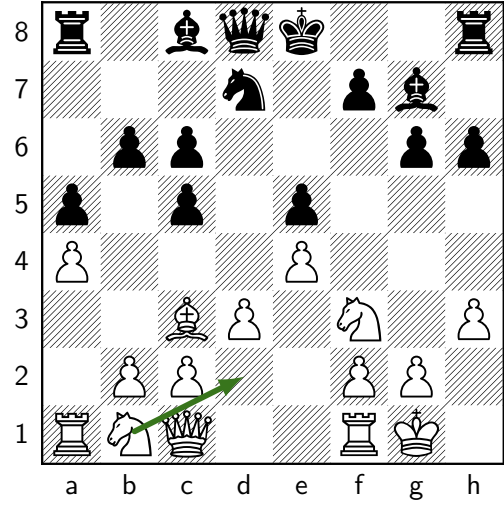


(d)

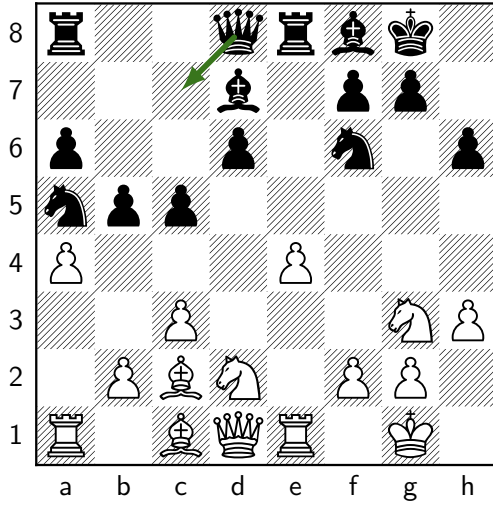
Figure 6: **Move 13, White: Kf1**. (a) Our position; Melanie plays Kf1 (g1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3631` (ply 27) [structural distance $d=69$, QAP obj=78]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra8*, our *Pd3* as their *Qd8*; transferred move g1f1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1352` (ply 18) [structural distance $d=80$, QAP obj=61]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *pf5* as their *Ra1*, our *pb5* as their *Bc1*; transferred move g1f1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1353` (ply 18) [structural distance $d=80$, QAP obj=61]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *pf5* as their *Ra1*, our *pb5* as their *Bc1*; transferred move g1f1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf1 (summed QAP objective 200, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



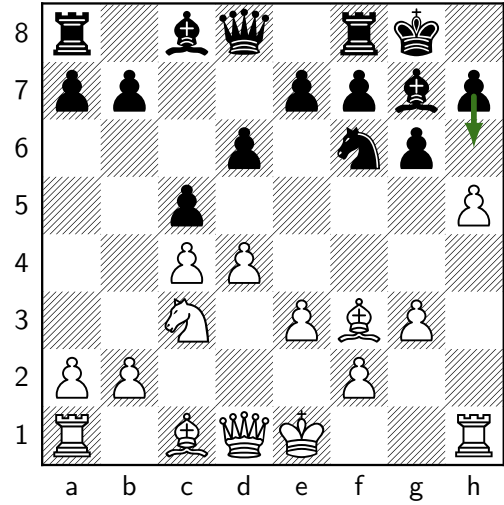
(a)



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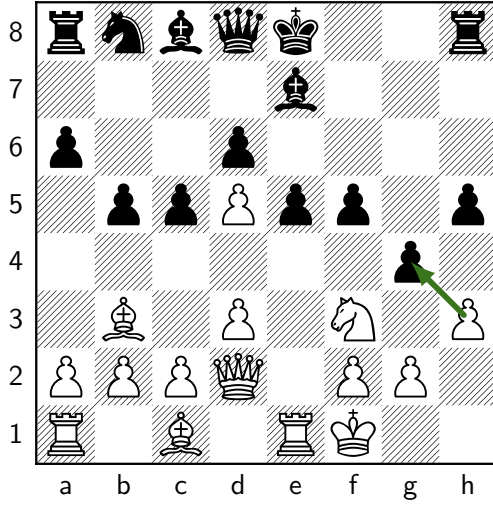


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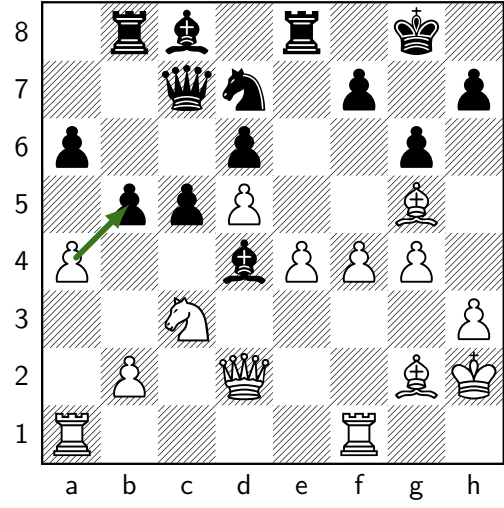


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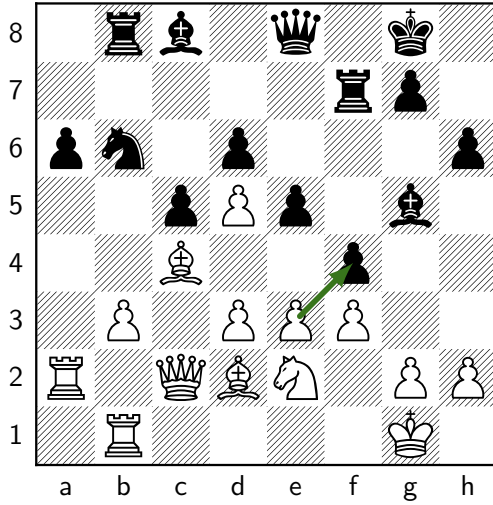
Figure 7: **Move 14, White: Qd2.** (a) Our position; Melanie plays Qd2 (d1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2518` (ply 22) [structural distance $d=84$, QAP obj=76]: the master played Nbd2; under μ our Qd1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our pd6 as their Ra1, our Bc1 as their Qc1; transferred move d1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#694` (ply 31) [structural distance $d=81$, QAP obj=73]: the master played Qc7; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf2 as their Bf8, our Nf3 as their Bd7; transferred move d1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#124` (ply 19) [structural distance $d=69$, QAP obj=49]: the master played h6; under μ our Qd1 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Ra8, our pd6 as their Bc8; transferred move d1d2 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Qd2 (summed QAP objective 334, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



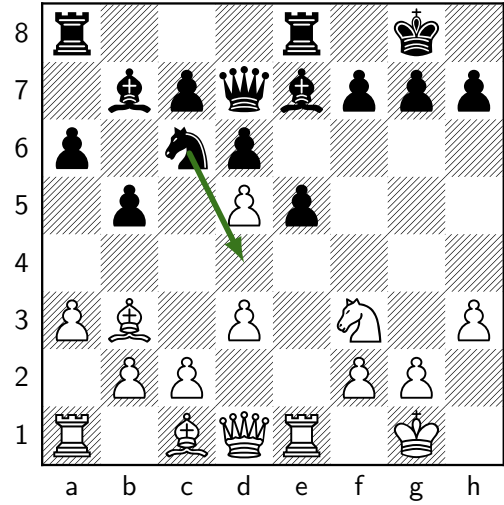
(a)



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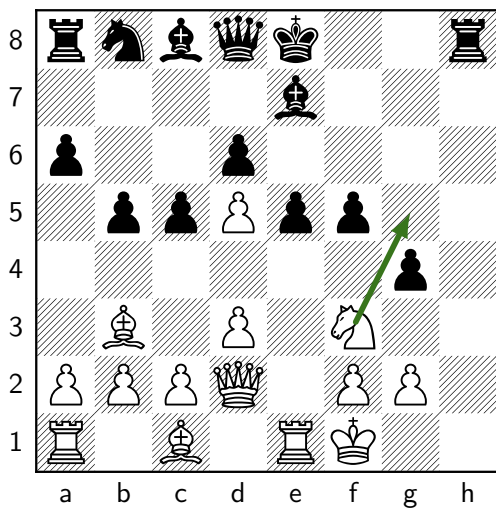


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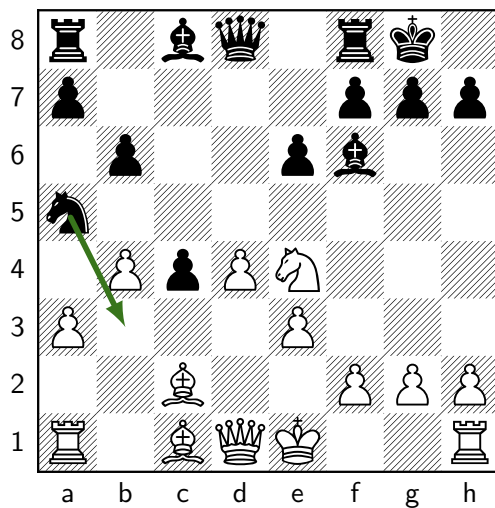


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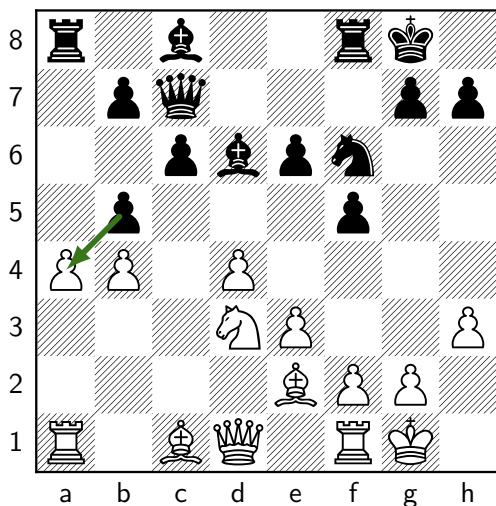
Figure 8: **Move 15, White: hxg4.** (a) Our position; Melanie plays **hxg4** (**h3g4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3679` (ply 44) [structural distance $d=98$, QAP obj=45]: the master played **axb5**; under μ our *Ph3* plays the role of that game’s *Pa4*; cross-type roles: our *Pf2* as their *Ra1*, our *Pc2* as their *Rf1*; transferred move **h3g4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1679` (ply 38) [structural distance $d=100$, QAP obj=39]: the master played **exf4**; under μ our *Ph3* plays the role of that game’s *Pe3*; cross-type roles: our *Pd5* as their *Rb1*, our *Pg2* as their *Bd2*; transferred move **h3g4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2079` (ply 25) [structural distance $d=101$, QAP obj=54]: the master played **Nd4**; under μ our *Nf3* plays the role of that game’s *Nc3*; cross-type roles: our *Pd5* as their *Ra8*, our *qd8* as their *Re8*; transferred move **f3e5**. Synthesis: 2 retrieved analogs transferred to **hxg4** (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



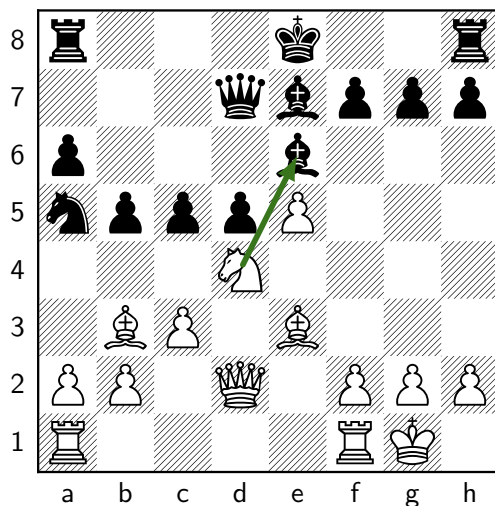
(a)



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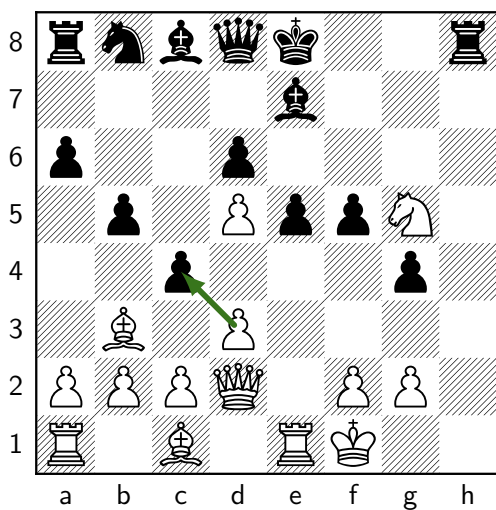


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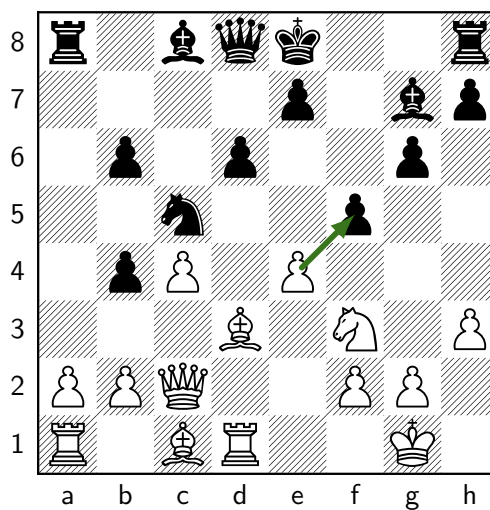


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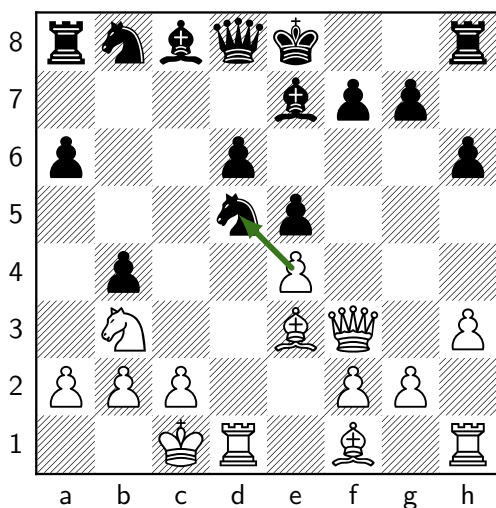
Figure 9: **Move 16, White: Ng5.** (a) Our position; Melanie plays Ng5 (f3g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#349` (ply 27) [structural distance $d=81$, QAP obj=53]: the master played Nb3; under μ our Nf3 plays the role of that game’s Na4; cross-type roles: our Pb2 as their Ra8, our Pd5 as their Bc8; transferred move f3g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2308` (ply 27) [structural distance $d=78$, QAP obj=36]: the master played bxa4; under μ our Nf3 plays the role of that game’s Pb4 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Rf8, our pg4 as their Qc7; transferred move f3g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1297` (ply 26) [structural distance $d=77$, QAP obj=43]: the master played Nxe6; under μ our Nf3 plays the role of that game’s Nd4; cross-type roles: our Pg2 as their Ra1, our Bc1 as their Pa2; transferred move f3d4. Synthesis: 2 retrieved analogs transferred to Ng5 (summed QAP objective 89, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



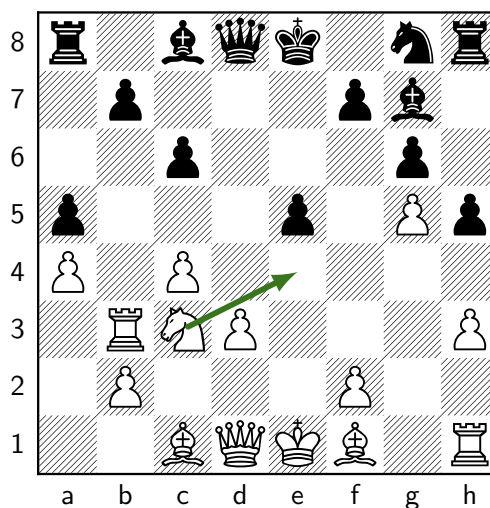
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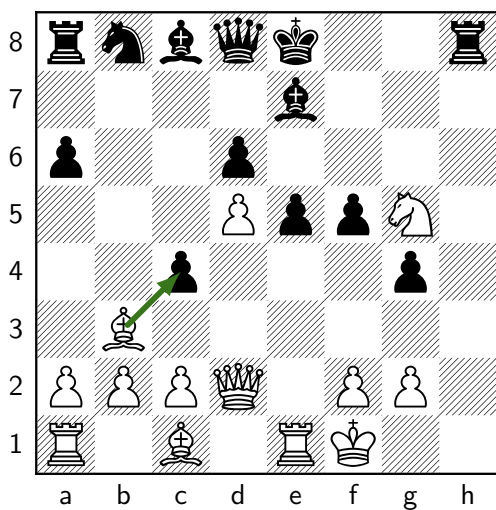


(c)

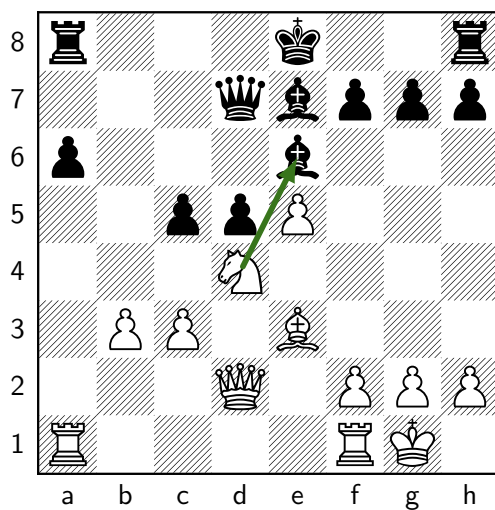


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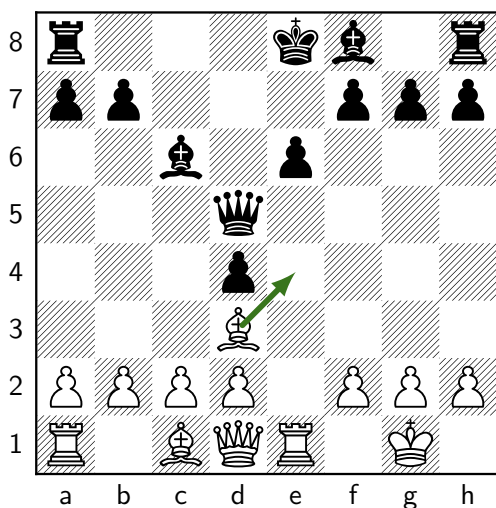
Figure 10: **Move 17, White: dxc4**. (a) Our position; Melanie plays dxc4 (d3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#1702 (ply 34) [structural distance $d=86$, QAP obj=43]: the master played **exf5**; under μ our $Pd3$ plays the role of that game’s $Pe4$; cross-type roles: our $Pa2$ as their $Ra1$, our $Pg2$ as their $Rd1$; transferred move **d3c4 (endorses this move)**. (c) Retrieved from ccr1_top_p0000#409 (ply 24) [structural distance $d=87$, QAP obj=40]: the master played **exd5**; under μ our $Pd3$ plays the role of that game’s $Pe4$; cross-type roles: our $Pd3$ as their $Rd1$, our $Pc2$ as their $Bf1$; transferred move **d3c4 (endorses this move)**. (d) Retrieved from ccr1_top_p0001#2833 (ply 26) [structural distance $d=87$, QAP obj=23]: the master played **Ne4**; under μ our $Pd3$ plays the role of that game’s $Nc3$ (a **cross-type** correspondence); cross-type roles: our $Ng5$ as their $Bc1$, our $Pf2$ as their $Bf1$; transferred move **d3c4 (endorses this move)**. Synthesis: 3 retrieved analogs transferred to dxc4 (summed QAP objective 106, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



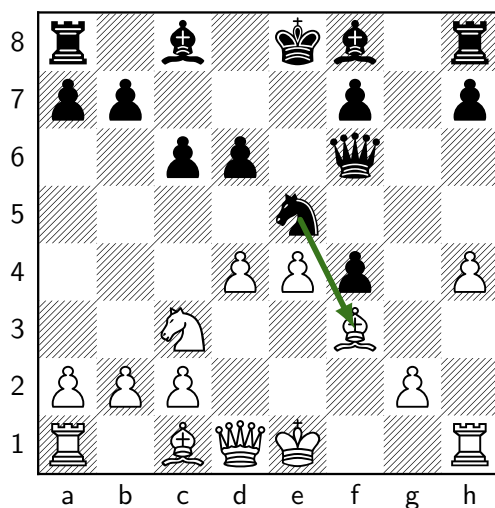
(a)



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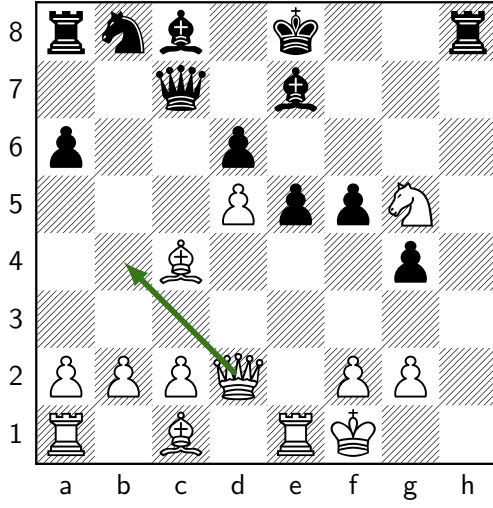


(c)

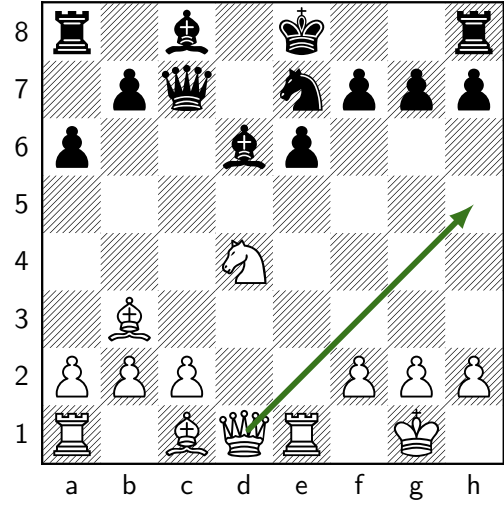


(d)

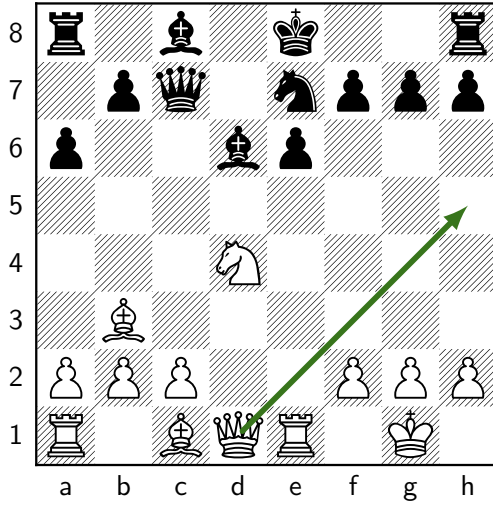
Figure 11: **Move 18, White: Bxc4.** (a) Our position; Melanie plays Bxc4 (b3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1050` (ply 32) [structural distance $d=86$, QAP obj=48]: the master played Nxe6; under μ our Bb3 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Rf1, our Pa2 as their Qd2; transferred move b3c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1592` (ply 22) [structural distance $d=89$, QAP obj=25]: the master played Be4; under μ our Bb3 plays the role of that game’s Bd3; cross-type roles: our bc8 as their Ra1, our Ra1 as their Bc1; transferred move b3c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#483` (ply 21) [structural distance $d=91$, QAP obj=24]: the master played Nxf3+; under μ our Bb3 plays the role of that game’s Ne4 (a **cross-type** correspondence); cross-type roles: our be7 as their Ra8, our Pa2 as their Bc8; transferred move b3c4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bxc4 (summed QAP objective 189, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



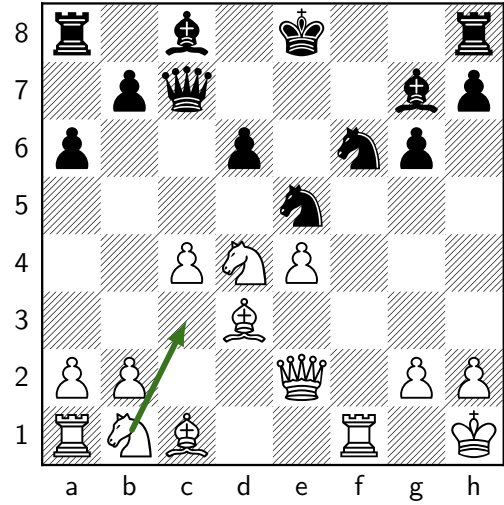
(a)



(b)

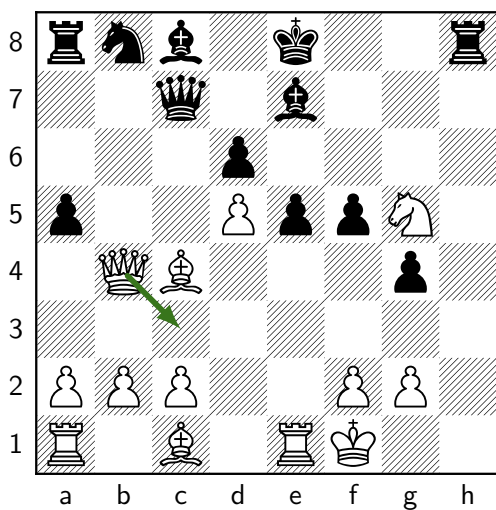


(c)

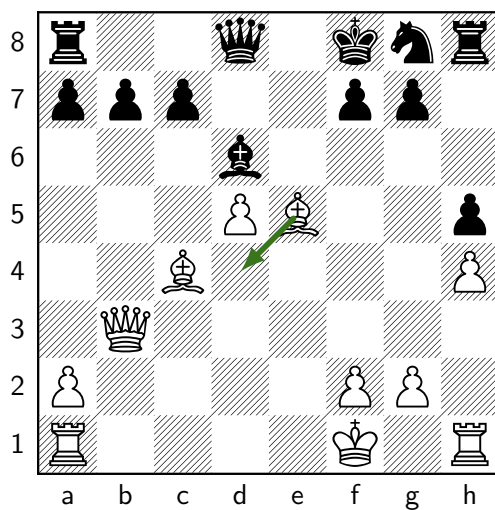


(d)

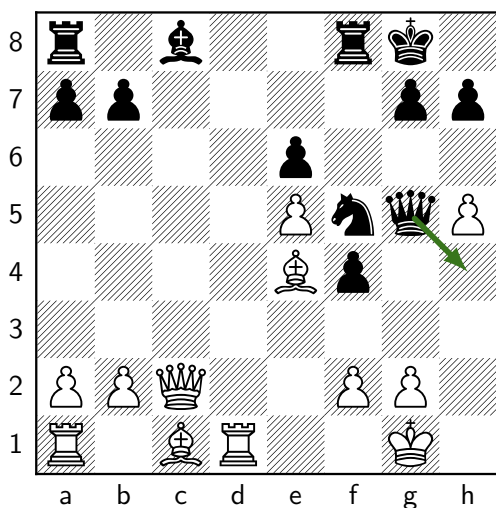
Figure 12: **Move 19, White: Qb4.** (a) Our position; Melanie plays **Qb4** (d2b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#665` (ply 24) [structural distance $d=86$, QAP obj=33]: the master played **Qh5**; under μ our *Qd2* plays the role of that game’s *Qd1*; cross-type roles: our *Pf2* as their *Ra1*, our *Ra1* as their *Pg2*; transferred move d2b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#666` (ply 24) [structural distance $d=86$, QAP obj=33]: the master played **Qh5**; under μ our *Qd2* plays the role of that game’s *Qd1*; cross-type roles: our *Pf2* as their *Ra1*, our *Ra1* as their *Pg2*; transferred move d2b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2413` (ply 26) [structural distance $d=92$, QAP obj=47]: the master played **Nc3**; under μ our *Re1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Bc1* as their *Ra1*, our *qc7* as their *Rf1*; transferred move e1e4. Synthesis: 2 retrieved analogs transferred to **Qb4** (summed QAP objective 66, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



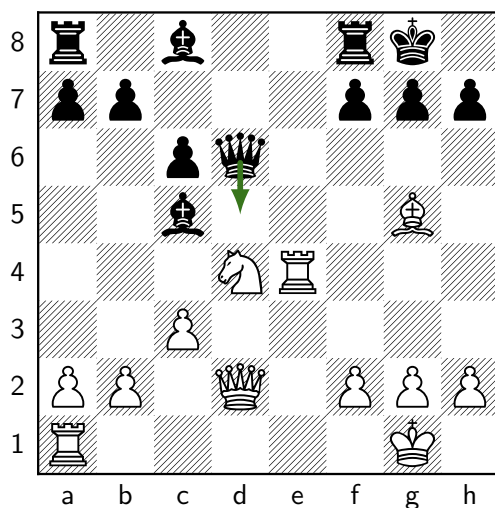
(a)



(b)

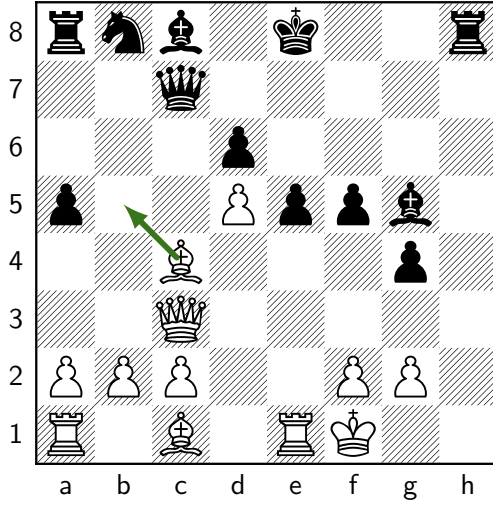


(c)

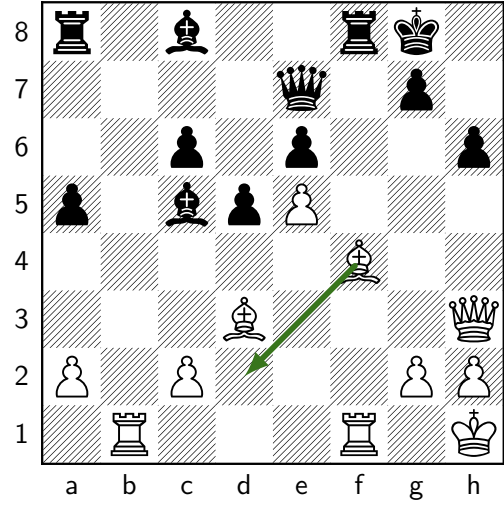


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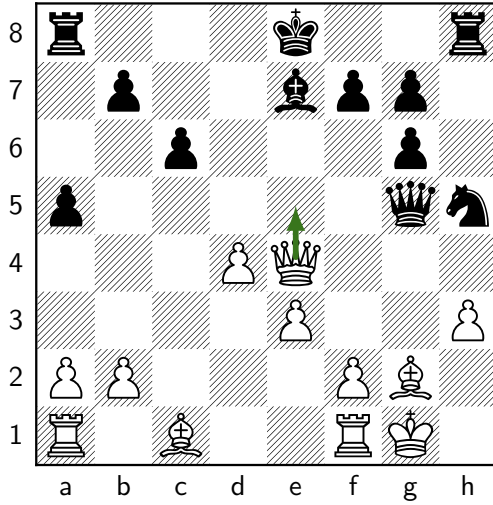
Figure 13: **Move 20, White: Qc3.** (a) Our position; Melanie plays Qc3 (b4c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3863` (ply 28) [structural distance $d=104$, QAP obj=54]: the master played Bd4; under μ our Qb4 plays the role of that game’s Be5 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Ra1, our Ra1 as their Pf2; transferred move b4c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2197` (ply 43) [structural distance $d=106$, QAP obj=44]: the master played Qh4; under μ our Qb4 plays the role of that game’s Qg4; cross-type roles: our Pg2 as their Ra8, our bc8 as their Rf8; transferred move b4c3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1138` (ply 33) [structural distance $d=102$, QAP obj=15]: the master played Qd5; under μ our Qb4 plays the role of that game’s Qd3; cross-type roles: our bc8 as their Ra8, our Pd5 as their Bc8; transferred move b4c3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qc3 (summed QAP objective 114, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



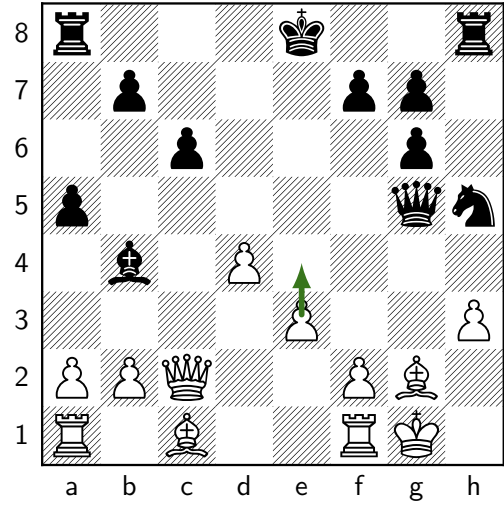
(a)



(b)

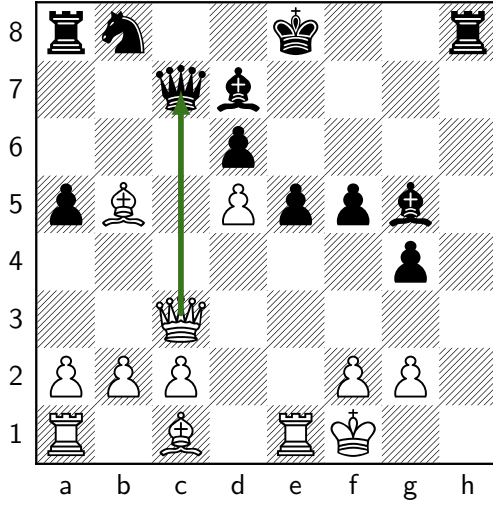


(c)

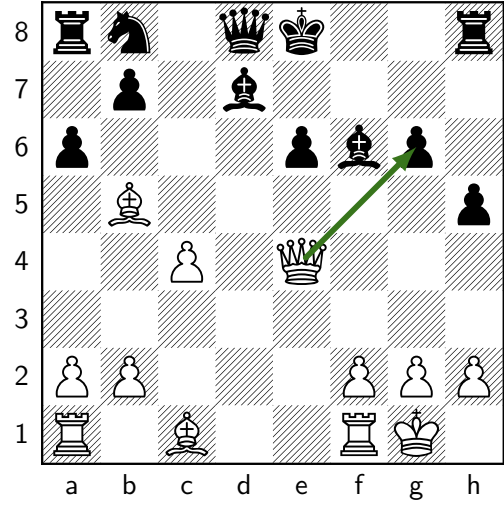


(d)

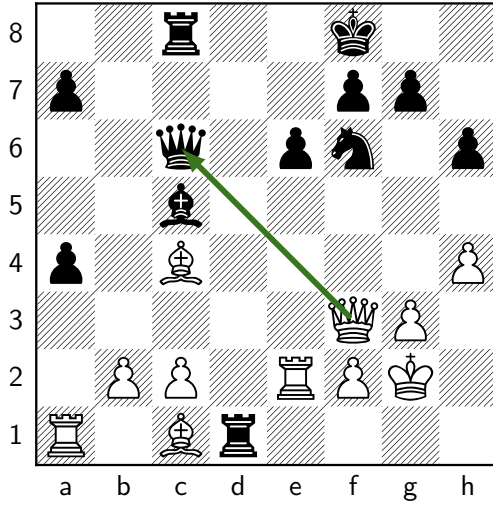
Figure 14: **Move 21, White: Bb5+.** (a) Our position; Melanie plays Bb5+ (c4b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2460` (ply 54) [structural distance $d=89$, QAP obj=53]: the master played Bd2; under μ our Bc4 plays the role of that game’s Bf4; cross-type roles: our Pd5 as their Rb1, our Pf2 as their Rf1; transferred move c4b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2019` (ply 38) [structural distance $d=95$, QAP obj=98]: the master played Qe5; under μ our Qc3 plays the role of that game’s Qe4; cross-type roles: our Pg2 as their Ra1, our Ra1 as their Pa2; transferred move c3e5. (d) Retrieved from `ccr1_top_p0001#2017` (ply 36) [structural distance $d=89$, QAP obj=70]: the master played e4; under μ our Bc4 plays the role of that game’s Pe3 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Ra1, our pf5 as their Bc1; transferred move c4d3. Synthesis: 1 retrieved analog transferred to Bb5+ (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



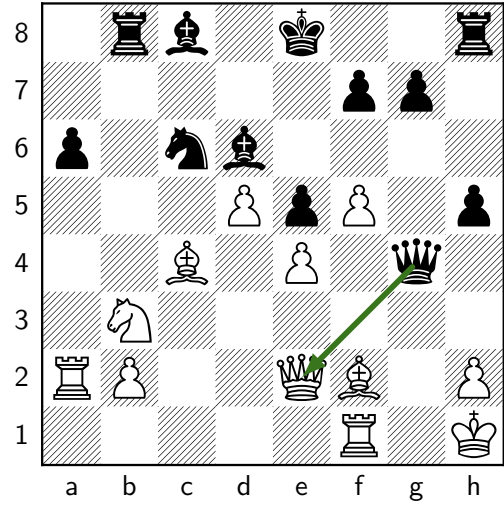
(a)



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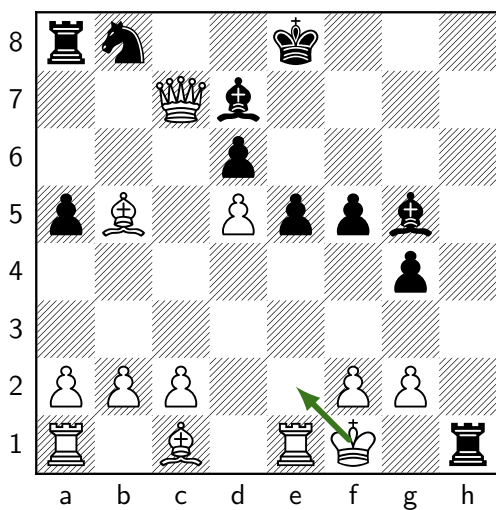


(c)

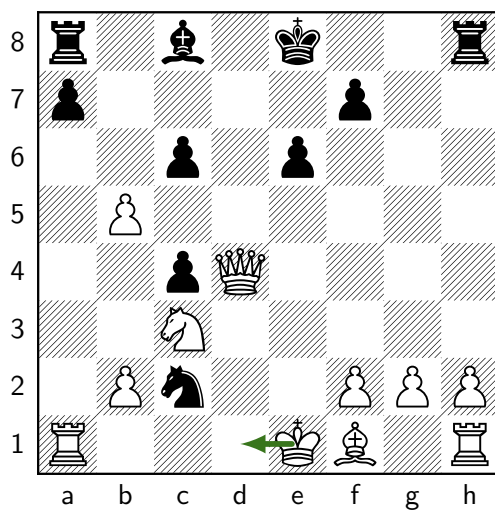


(d)

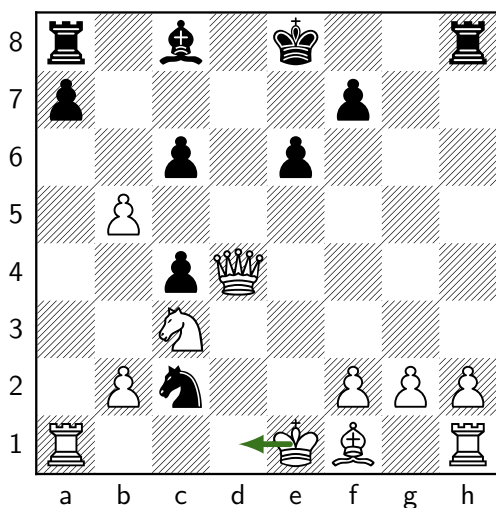
Figure 15: **Move 22, White: Qxc7**. (a) Our position; Melanie plays Qxc7 (c3c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2599` (ply 30) [structural distance $d=91$, QAP obj=74]: the master played Qxg6+; under μ our Qc3 plays the role of that game's Qe4; cross-type roles: our pf5 as their Bc1, our bg5 as their Pf2; transferred move c3c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#83` (ply 46) [structural distance $d=90$, QAP obj=59]: the master played Qxc6; under μ our Qc3 plays the role of that game's Qf3; cross-type roles: our Pf2 as their Ra1, our Bb5 as their rd1; transferred move c3c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2815` (ply 47) [structural distance $d=83$, QAP obj=52]: the master played Qxe2; under μ our Qc3 plays the role of that game's Qg5; cross-type roles: our bg5 as their Rb8, our Ra1 as their Pf7; transferred move c3c7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qxc7 (summed QAP objective 185, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



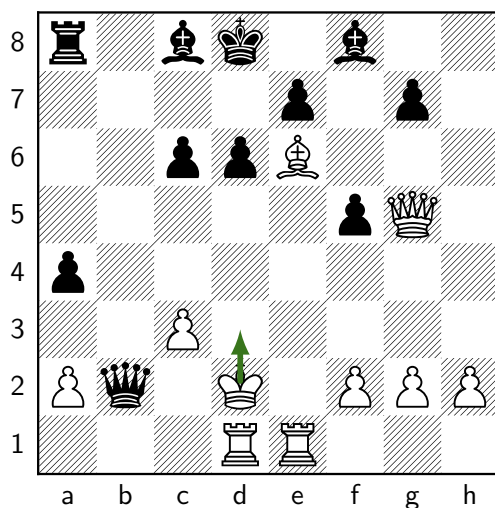
(a)



(b)

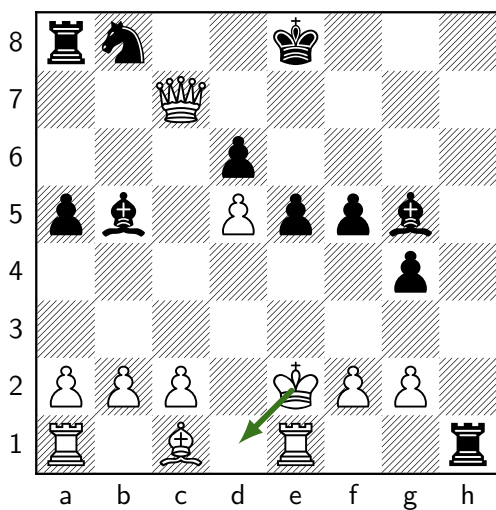


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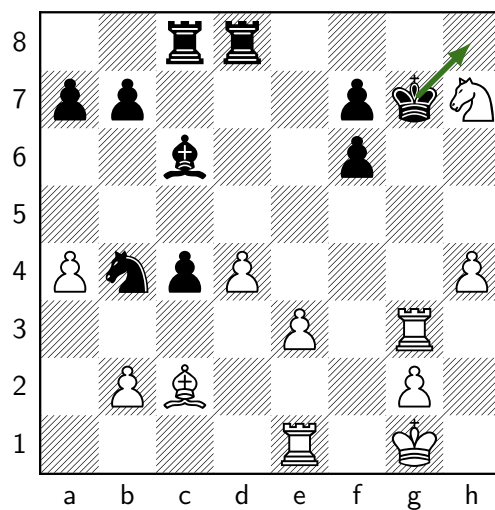


(d)

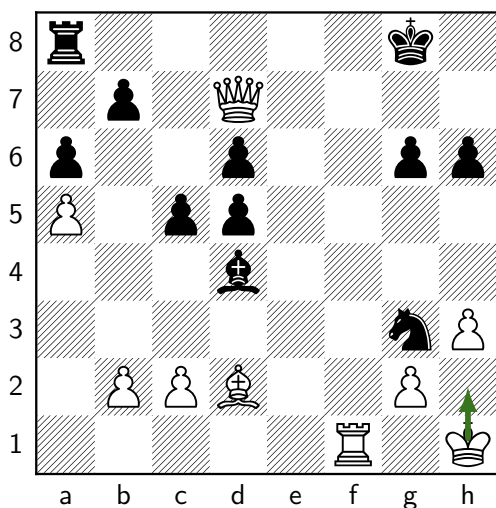
Figure 16: **Move 23, White: Ke2.** (a) Our position; Melanie plays Ke2 (f1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2868` (ply 32) [structural distance $d=120$, QAP obj=83]: the master played Kd1; under μ our *Kf1* plays the role of that game’s *Ke1*; cross-type roles: our *Bb5* as their *Ra1*, our *Pf2* as their *Bf1*; transferred move f1e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3901` (ply 32) [structural distance $d=120$, QAP obj=83]: the master played Kd1; under μ our *Kf1* plays the role of that game’s *Ke1*; cross-type roles: our *Bb5* as their *Ra1*, our *Pf2* as their *Bf1*; transferred move f1e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2537` (ply 42) [structural distance $d=120$, QAP obj=83]: the master played Kd3; under μ our *Kf1* plays the role of that game’s *Kd2*; cross-type roles: our *Pg2* as their *Re1*, our *bg5* as their *Pa2*; transferred move f1e2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ke2 (summed QAP objective 393, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



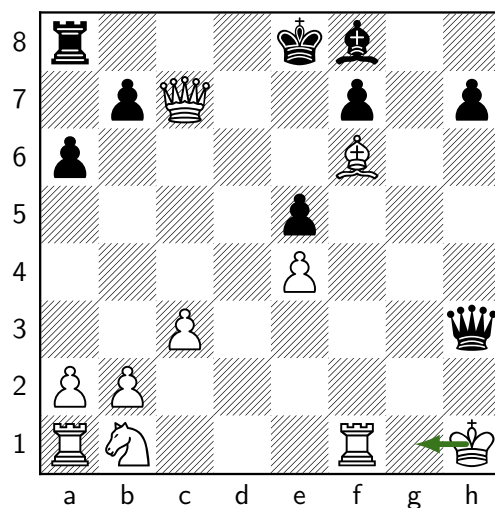
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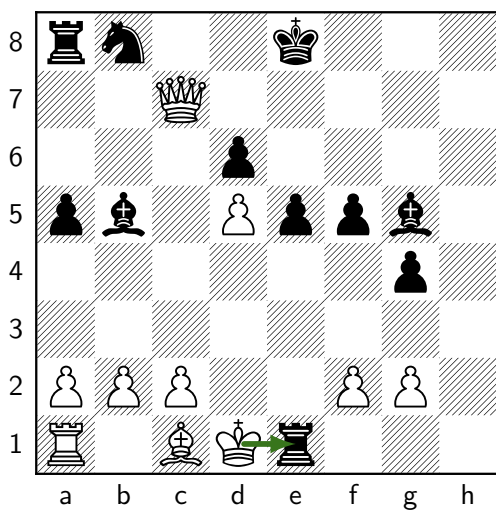


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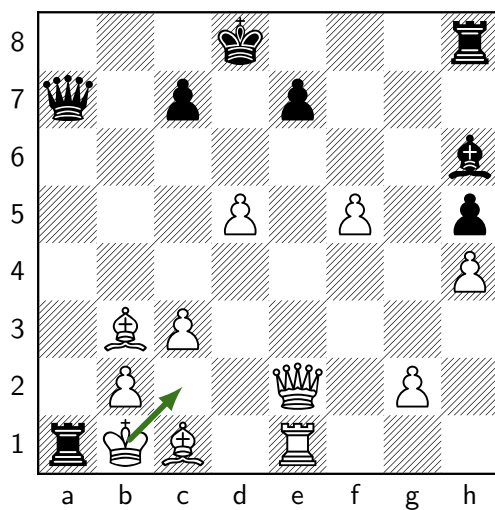


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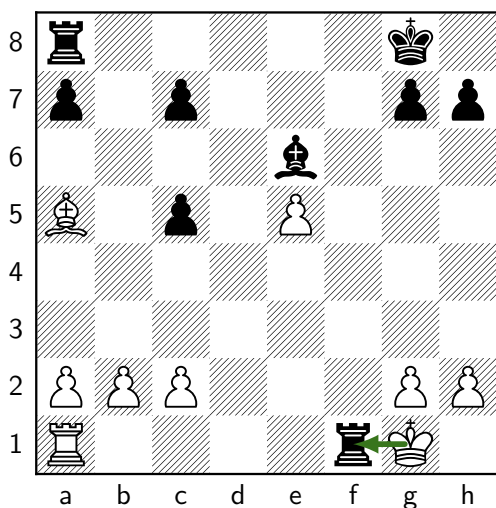
Figure 17: **Move 24, White: Kd1**. (a) Our position; Melanie plays Kd1 (e2d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#293` (ply 59) [structural distance $d=99$, QAP obj=54]: the master played Kh8; under μ our Ke2 plays the role of that game’s Kg2; cross-type roles: our Bc1 as their Rd8, our bg5 as their nh7; transferred move e2d1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1109` (ply 56) [structural distance $d=89$, QAP obj=50]: the master played Kh2; under μ our Ke2 plays the role of that game’s Kh1; cross-type roles: our Ra1 as their Pb2, our Pf2 as their Bd2; transferred move e2d1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#737` (ply 36) [structural distance $d=100$, QAP obj=49]: the master played Kg1; under μ our Ke2 plays the role of that game’s Kh1; cross-type roles: our pe5 as their Nb1, our Bc1 as their Pc3; transferred move e2d1 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Kd1 (summed QAP objective 446, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



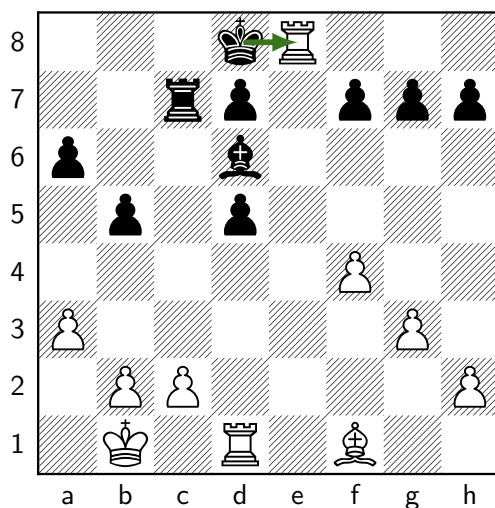
(a)



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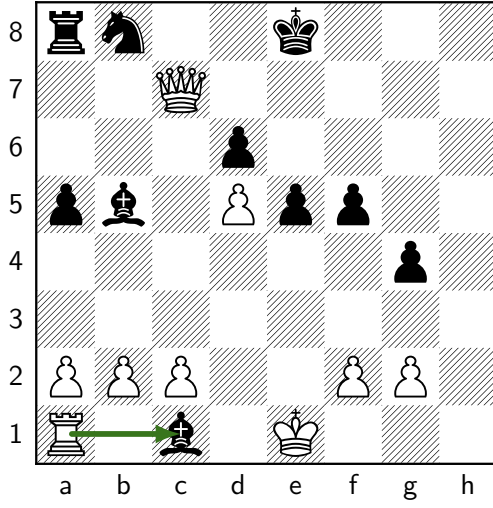


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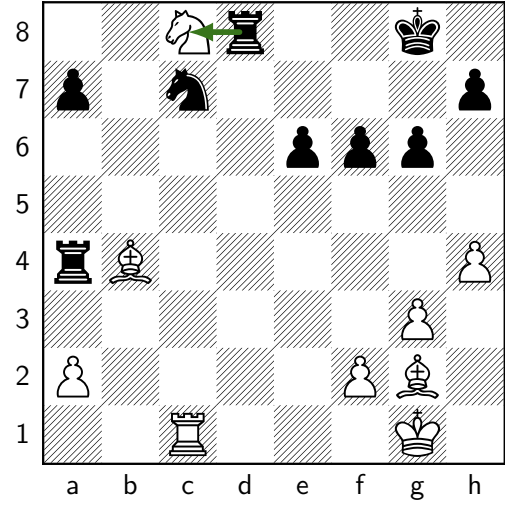


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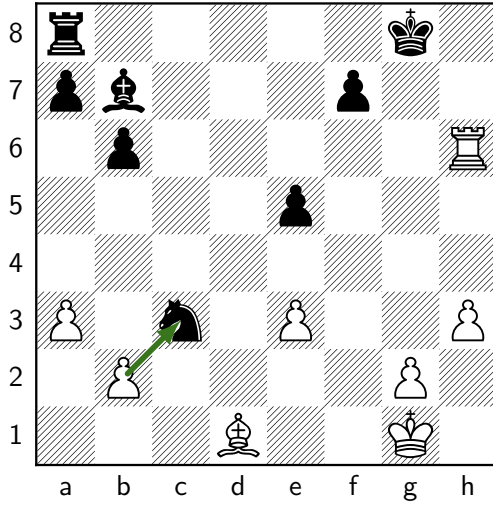
Figure 18: **Move 25, White: Kxe1**. (a) Our position; Melanie plays Kxe1 (d1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1850` (ply 64) [structural distance $d=76$, QAP obj=74]: the master played Kc2; under μ our Kd1 plays the role of that game’s Kb1; cross-type roles: our Pf2 as their Bb3, our ra8 as their qa7; transferred move d1e1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2071` (ply 36) [structural distance $d=86$, QAP obj=71]: the master played Kxf1; under μ our Kd1 plays the role of that game’s Kg1; cross-type roles: our Bc1 as their Ph2, our Pg2 as their Ba5; transferred move d1e1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2071` (ply 41) [structural distance $d=86$, QAP obj=71]: the master played Kxe8; under μ our Kd1 plays the role of that game’s Kd1; cross-type roles: our Bc1 as their Rc7, our Ra1 as their Pd7; transferred move d1e1 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Kxe1 (summed QAP objective 817, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



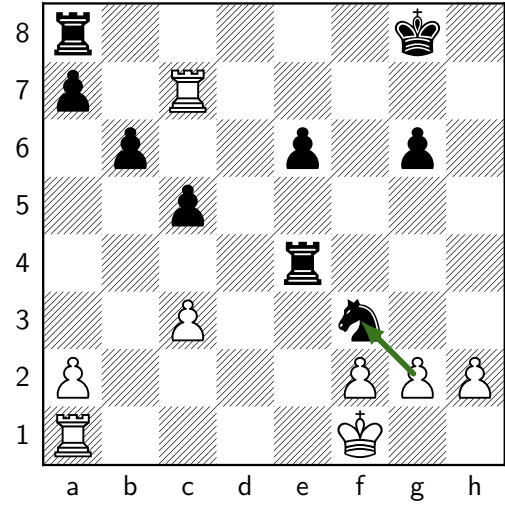
(a)



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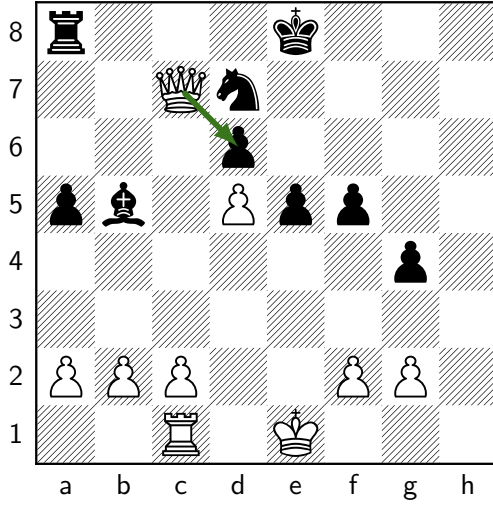


(c)

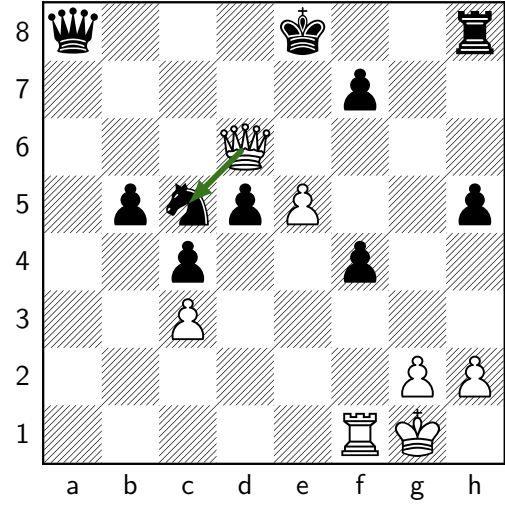


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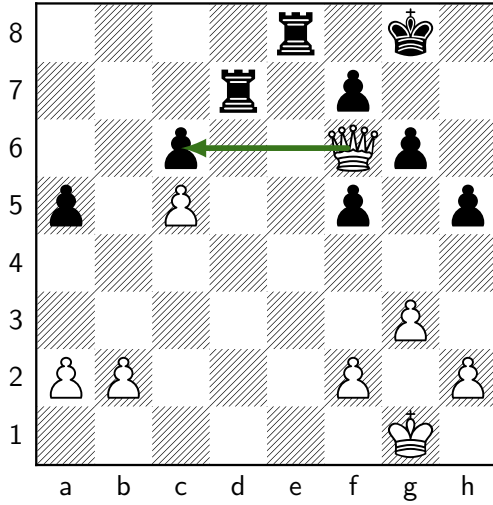
Figure 19: **Move 26, White: Rxc1**. (a) Our position; Melanie plays Rxc1 (a1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2961` (ply 63) [structural distance $d=63$, QAP obj=38]: the master played Rxc8; under μ our *Ra1* plays the role of that game's *Rd1*; cross-type roles: our *b1* as their *nc8*, our *Pc2* as their *Nc7*; transferred move a1c1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2130` (ply 58) [structural distance $d=65$, QAP obj=29]: the master played bxc3; under μ our *Ra1* plays the role of that game's *Pb2* (a **cross-type** correspondence); cross-type roles: our *Pb2* as their *Bd1*, our *b1* as their *nc3*; transferred move a1c1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#154` (ply 42) [structural distance $d=65$, QAP obj=28]: the master played gxf3; under μ our *Ra1* plays the role of that game's *Pg2* (a **cross-type** correspondence); cross-type roles: our *Pg2* as their *Ra1*, our *b1* as their *nf3*; transferred move a1c1 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Rxc1 (summed QAP objective 172, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



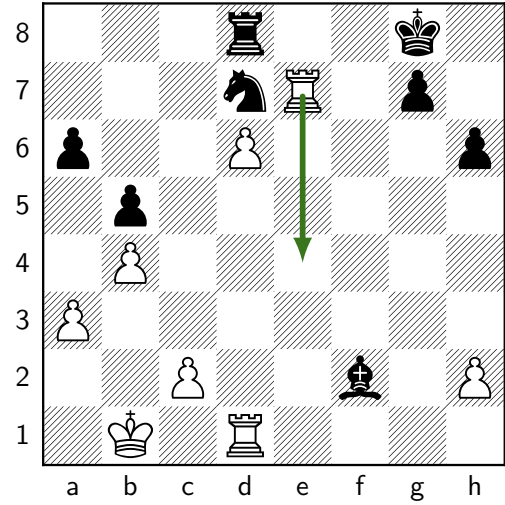
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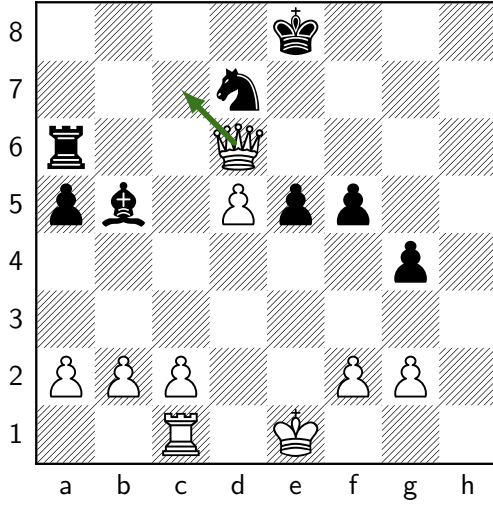


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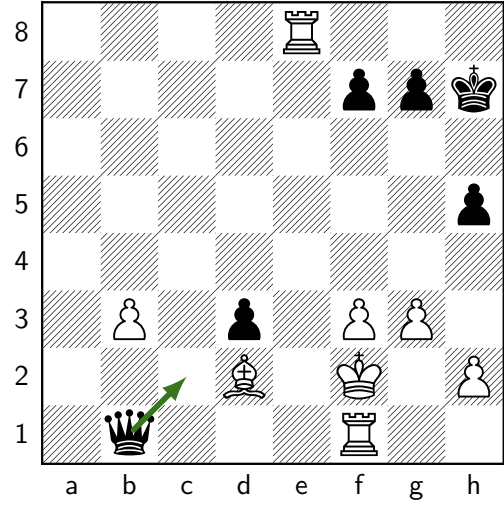


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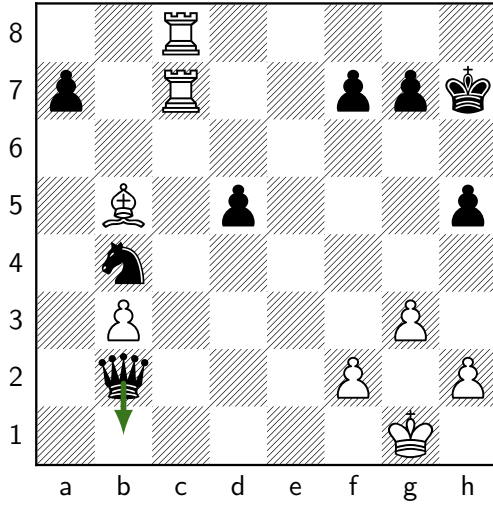
Figure 20: **Move 27, White: Qxd6.** (a) Our position; Melanie plays Qxd6 (c7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2113` (ply 56) [structural distance $d=56$, QAP obj=41]: the master played Qxc5; under μ our Qc7 plays the role of that game's Qd6; cross-type roles: our bb5 as their pf7, our ra8 as their qa8; transferred move c7d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1317` (ply 56) [structural distance $d=57$, QAP obj=34]: the master played Qxc6; under μ our Qc7 plays the role of that game's Qf6; cross-type roles: our Rc1 as their Pa2, our nd7 as their pf5; transferred move c7d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1710` (ply 68) [structural distance $d=55$, QAP obj=29]: the master played Re4; under μ our Qc7 plays the role of that game's Re7 (a **cross-type** correspondence); transferred move c7d6 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Qxd6 (summed QAP objective 177, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



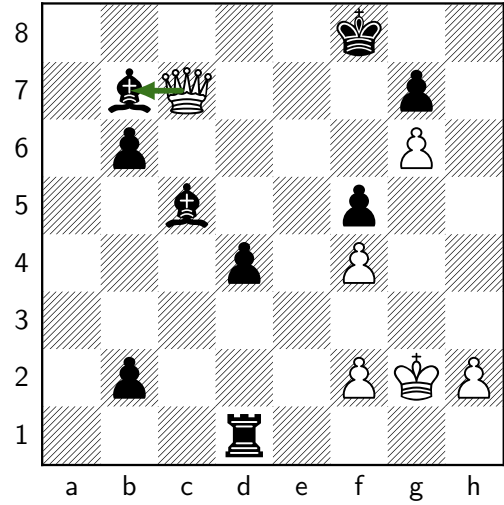
(a)



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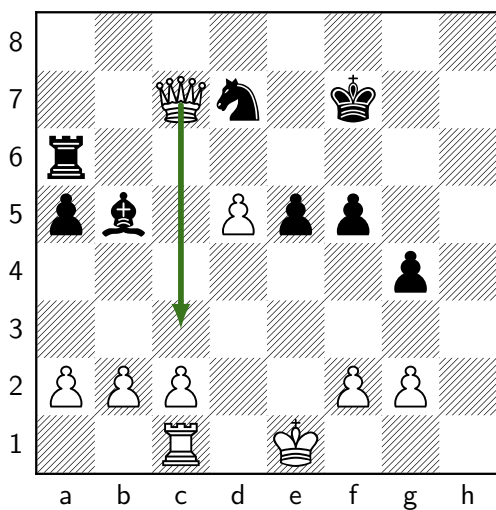


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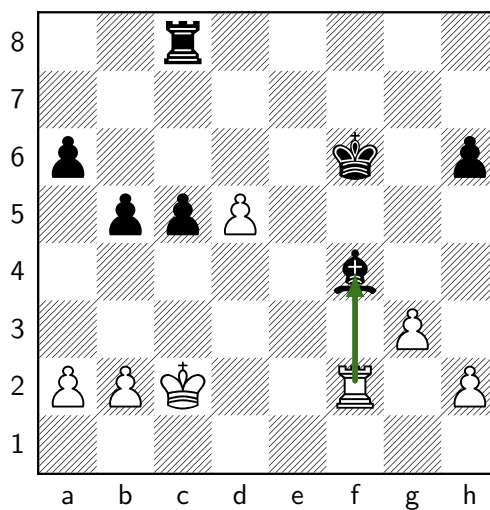


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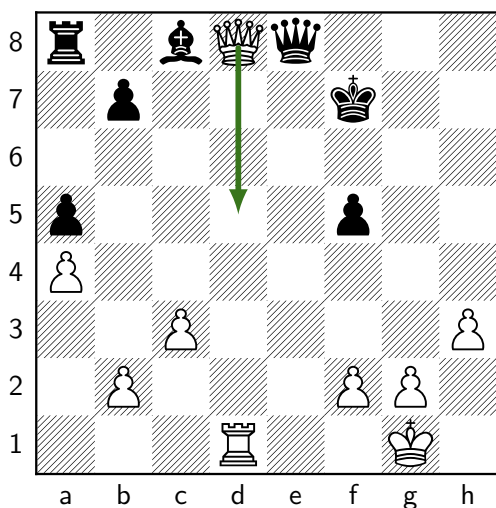
Figure 21: **Move 28, White: Qc7**. (a) Our position; Melanie plays Qc7 (d6c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2463` (ply 67) [structural distance $d=69$, QAP obj=24]: the master played Qc2; under μ our Qd6 plays the role of that game’s Qb8; cross-type roles: our bb5 as their re8, our Rc1 as their Pg7; transferred move d6c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3278` (ply 57) [structural distance $d=66$, QAP obj=8]: the master played Qb1+; under μ our Qd6 plays the role of that game’s Qb7; cross-type roles: our bb5 as their rc8, our pa5 as their rc7; transferred move d6c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2994` (ply 70) [structural distance $d=61$, QAP obj=4]: the master played Qxb7; under μ our Qd6 plays the role of that game’s Qc7; cross-type roles: our bb5 as their rd1, our Rc1 as their pb2; transferred move d6c7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qc7 (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



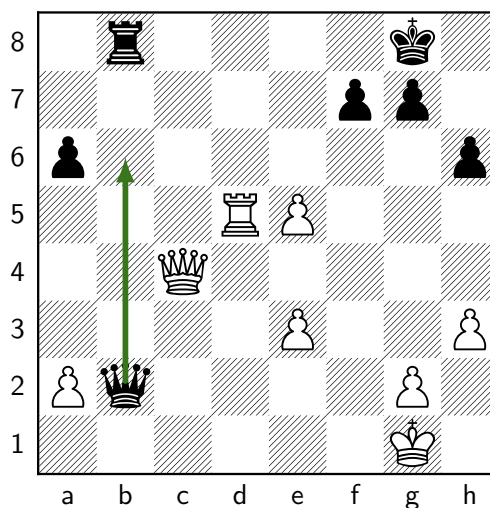
(a)



(b)

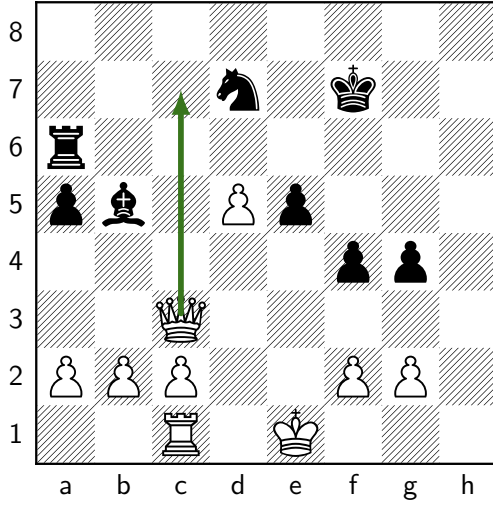


(c)

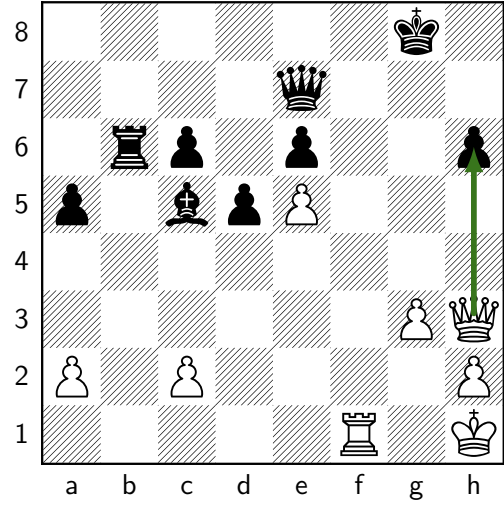


(d)

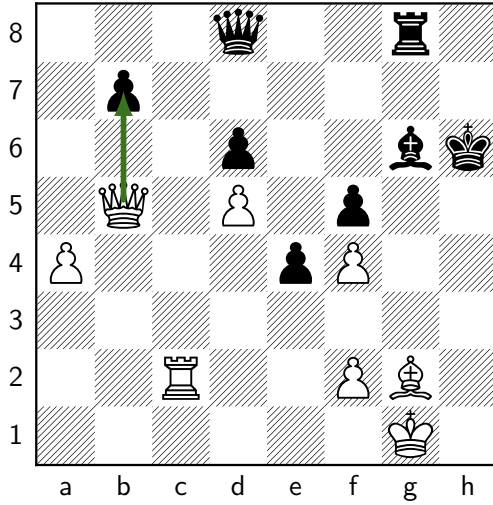
Figure 22: **Move 29, White: Qc3.** (a) Our position; Melanie plays Qc3 (c7c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2381` (ply 66) [structural distance $d=58$, QAP obj=37]: the master played $Rxf4+$; under μ our $Qc7$ plays the role of that game’s $Rf2$ (a **cross-type** correspondence); cross-type roles: our $Rc1$ as their $Pa2$, our $Qc7$ as their $Pg3$; transferred move c7c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3564` (ply 64) [structural distance $d=57$, QAP obj=37]: the master played $Qd5+$; under μ our $Qc7$ plays the role of that game’s $Qd8$; cross-type roles: our $Pc2$ as their $Rd1$, our $Rc1$ as their $Pa4$; transferred move c7c3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1881` (ply 53) [structural distance $d=60$, QAP obj=10]: the master played $Qb6$; under μ our $Qc7$ plays the role of that game’s $Qb7$; cross-type roles: our $bb5$ as their $qc4$, our $nd7$ as their $pa2$; transferred move c7c3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qc3 (summed QAP objective 90, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



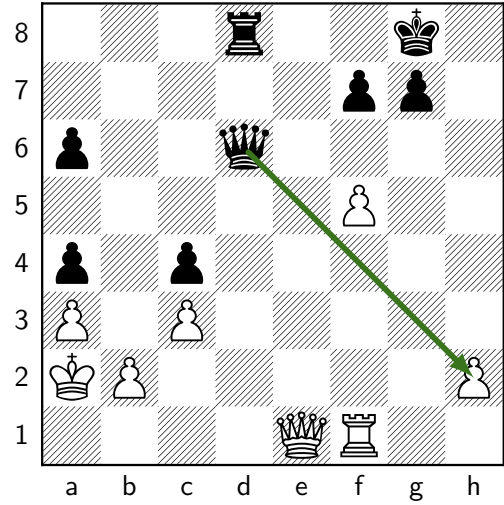
(a)



(b)

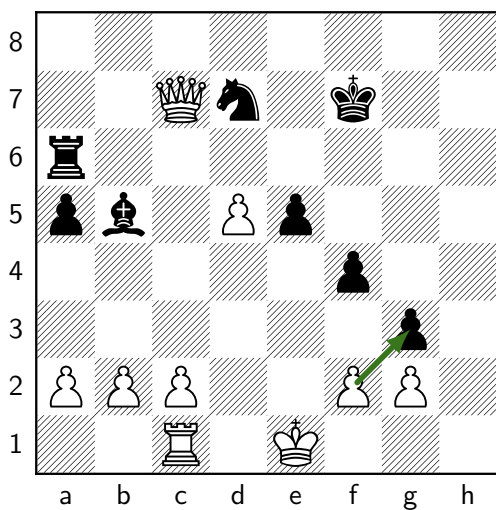


(c)

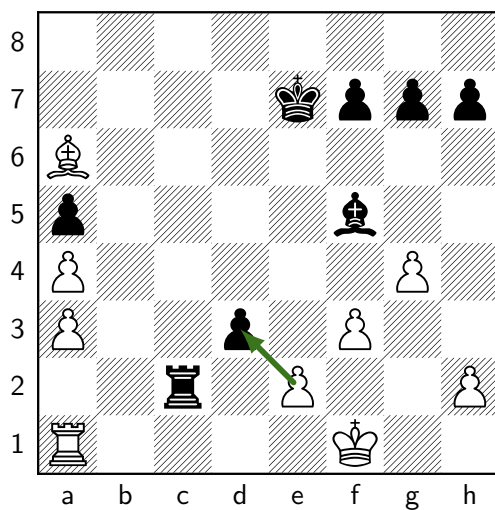


(d)

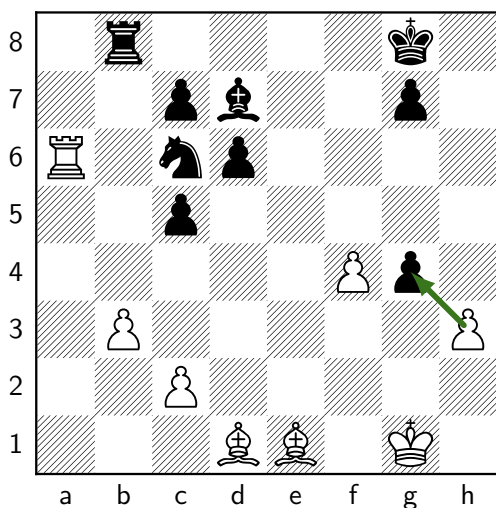
Figure 23: **Move 30, White: Qc7.** (a) Our position; Melanie plays Qc7 (c3c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2460` (ply 64) [structural distance $d=59$, QAP obj=37]: the master played Qxh6; under μ our Qc3 plays the role of that game’s Qh3; cross-type roles: our pf4 as their bc5, our bb5 as their pd5; transferred move c3c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1017` (ply 70) [structural distance $d=67$, QAP obj=36]: the master played Qxb7; under μ our Qc3 plays the role of that game’s Qb5; cross-type roles: our Pf2 as their Rc2, our Qc3 as their Bg2; transferred move c3c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1807` (ply 69) [structural distance $d=70$, QAP obj=28]: the master played Qxh2; under μ our Qc3 plays the role of that game’s Qd3; cross-type roles: our Pc2 as their Rd8, our Rc1 as their Pc4; transferred move c3c7 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Qc7 (summed QAP objective 191, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



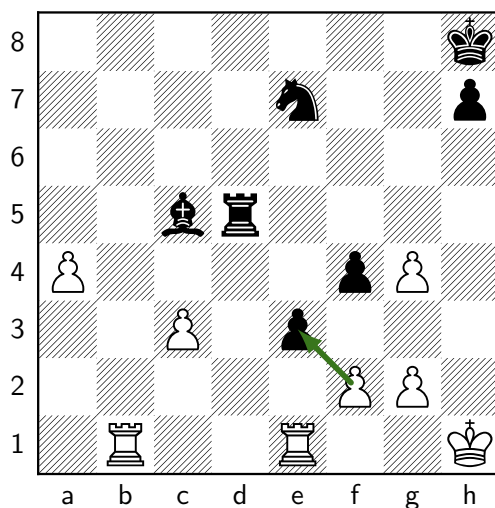
(a)



(b)

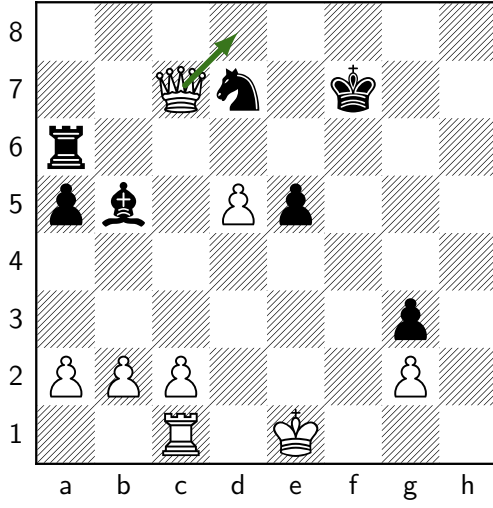


(c)

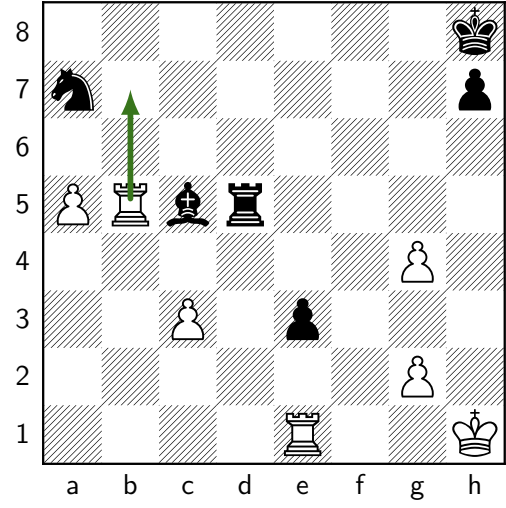


(d)

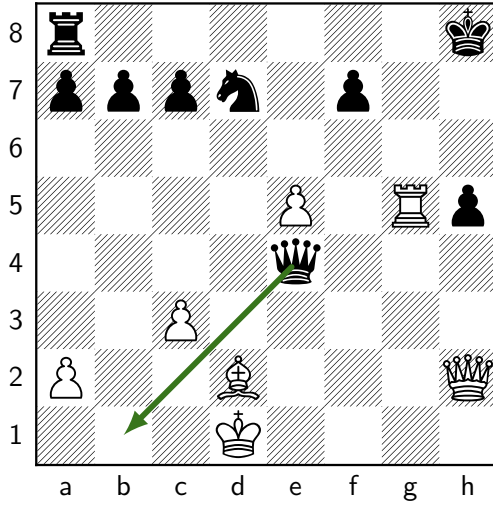
Figure 24: **Move 31, White: f2g3.** (a) Our position; Melanie plays f2g3. Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#1066 (ply 54) [structural distance $d=57$, QAP obj=34]: the master played exd3; under μ our Pf2 plays the role of that game’s Pe2; cross-type roles: our pf4 as their rc2, our ra6 as their Pa3; transferred move f2g3 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#891 (ply 60) [structural distance $d=63$, QAP obj=23]: the master played hxg4; under μ our Pf2 plays the role of that game’s Ph3; cross-type roles: our Pb2 as their Bd1, our Pg2 as their Be1; transferred move f2g3 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#1507 (ply 68) [structural distance $d=59$, QAP obj=16]: the master played fxe3; under μ our Pf2 plays the role of that game’s Pf2; cross-type roles: our Pc2 as their Rb1, our Qc7 as their Pg4; transferred move f2g3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to f2g3 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



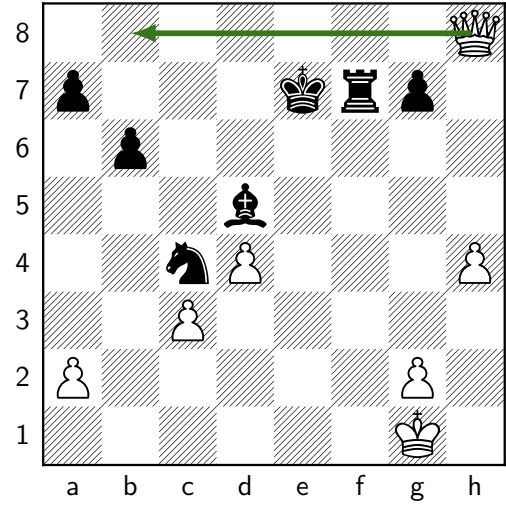
(a)



(b)

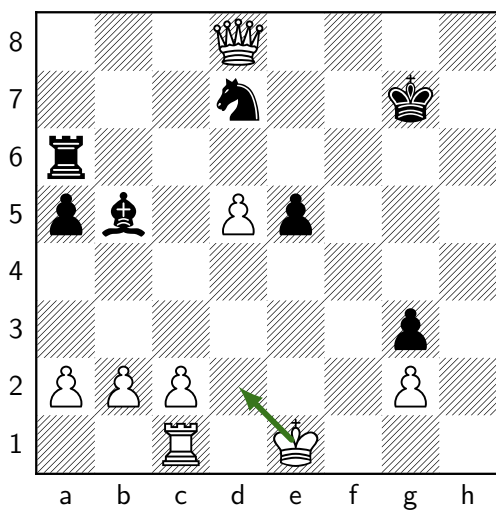


(c)

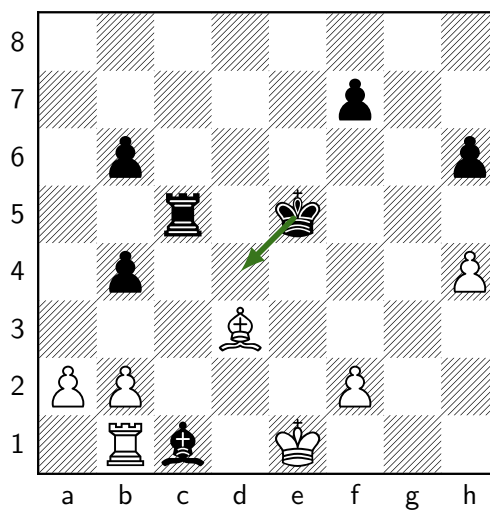


(d)

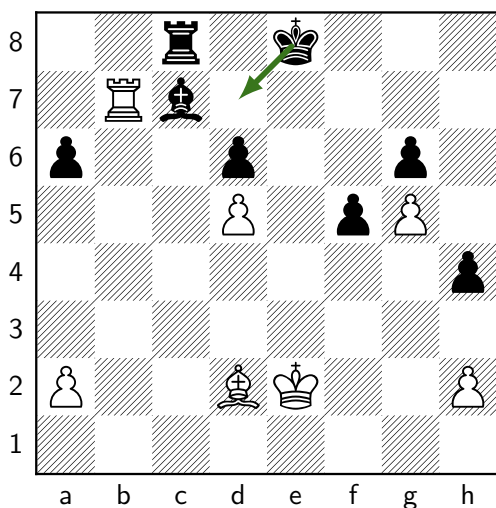
Figure 25: **Move 32, White: Qd8.** (a) Our position; Melanie plays Qd8 (c7d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1507` (ply 74) [structural distance $d=50$, QAP obj=43]: the master played Rb7; under μ our Qc7 plays the role of that game’s Rb5 (a **cross-type** correspondence); cross-type roles: our bb5 as their Pg2, our nd7 as their bc5; transferred move c7d8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1007` (ply 65) [structural distance $d=55$, QAP obj=9]: the master played Qb1+; under μ our Qc7 plays the role of that game’s Qe5; cross-type roles: our nd7 as their pe5, our bb5 as their pc3; transferred move c7d8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2006` (ply 72) [structural distance $d=40$, QAP obj=8]: the master played Qb8; under μ our Qc7 plays the role of that game’s Qh8; cross-type roles: our Rc1 as their Pa2, our Pc2 as their nc4; transferred move c7d8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qd8 (summed QAP objective 112, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



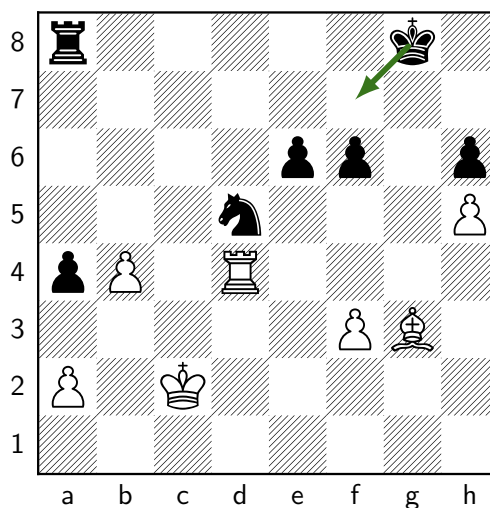
(a)



(b)

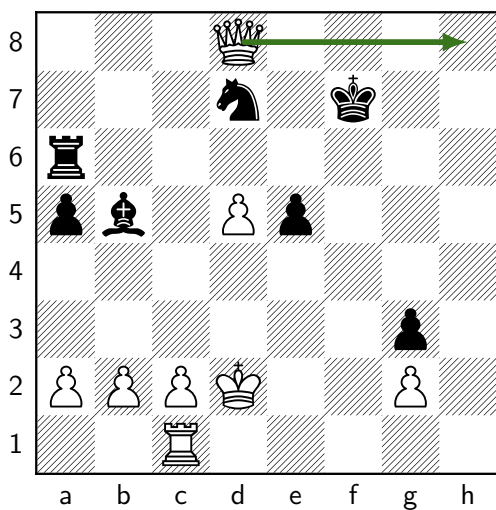


(c)

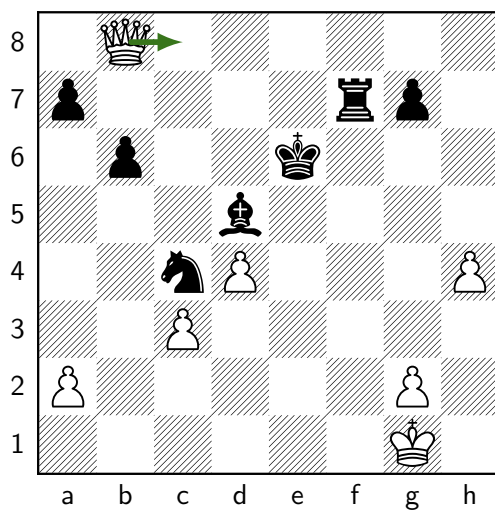


(d)

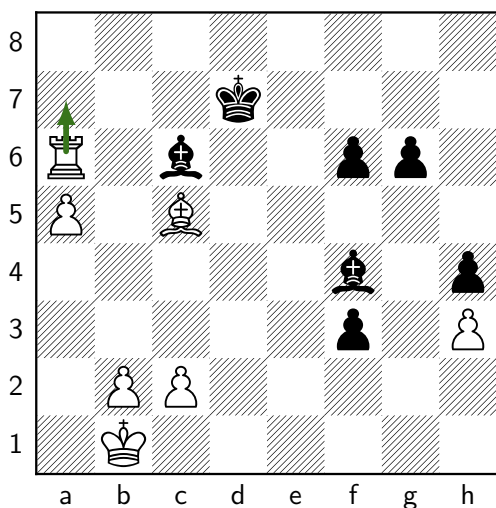
Figure 26: **Move 33, White: Kd2.** (a) Our position; Melanie plays Kd2 (e1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#445` (ply 73) [structural distance $d=46$, QAP obj=22]: the master played Kd4; under μ our *Ke1* plays the role of that game’s *Ke4*; cross-type roles: our *Qd8* as their *rb1*, our *nd7* as their *Bc1*; transferred move e1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#343` (ply 63) [structural distance $d=45$, QAP obj=22]: the master played Kd7; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd8* as their *rb7*, our *nd7* as their *Bc7*; transferred move e1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#540` (ply 67) [structural distance $d=47$, QAP obj=21]: the master played Kf7; under μ our *Ke1* plays the role of that game’s *Kg1*; cross-type roles: our *bb5* as their *Pe6*, our *ra6* as their *Ph6*; transferred move e1d2 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Kd2 (summed QAP objective 144, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



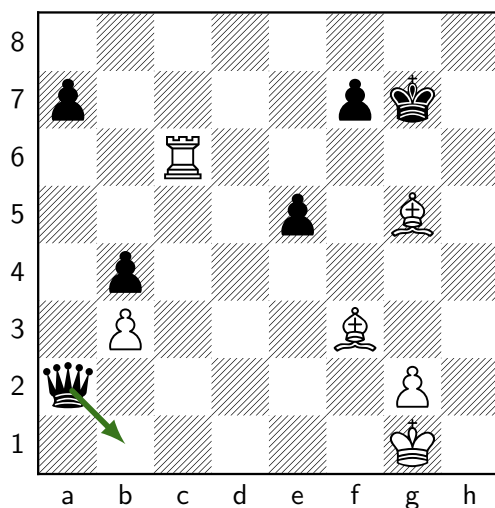
(a)



(b)

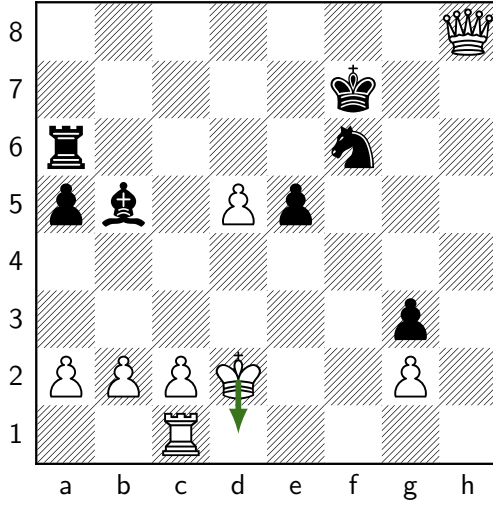


(c)

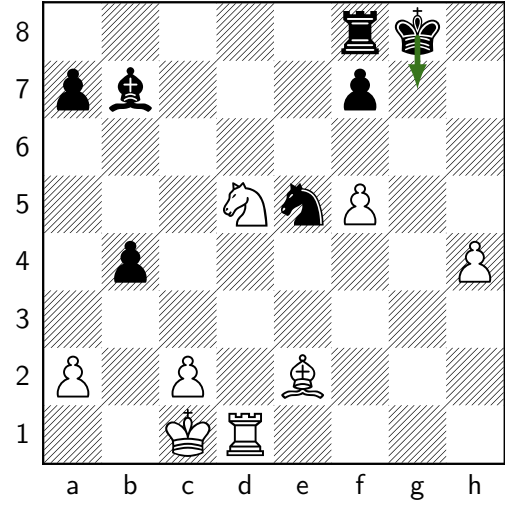


(d)

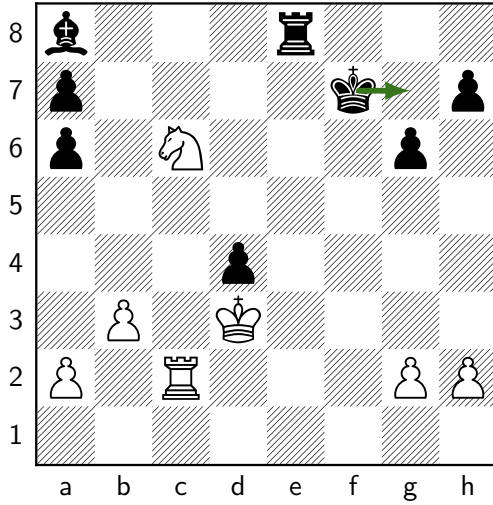
Figure 27: **Move 34, White: Qh8.** (a) Our position; Melanie plays Qh8 (d8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2006` (ply 74) [structural distance $d=46$, QAP obj=11]: the master played Qc8+; under μ our Qd8 plays the role of that game’s Qb8; cross-type roles: our Rc1 as their Pa2, our ra6 as their nc4; transferred move d8h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#693` (ply 74) [structural distance $d=46$, QAP obj=10]: the master played Ra7+; under μ our Qd8 plays the role of that game’s Ra6 (a **cross-type** correspondence); cross-type roles: our Rc1 as their Pb2, our Pd5 as their bf4; transferred move d8h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#980` (ply 67) [structural distance $d=41$, QAP obj=9]: the master played Qb1+; under μ our Qd8 plays the role of that game’s Qa7; cross-type roles: our Pg2 as their bg5, our nd7 as their pb3; transferred move d8h8 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qh8 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



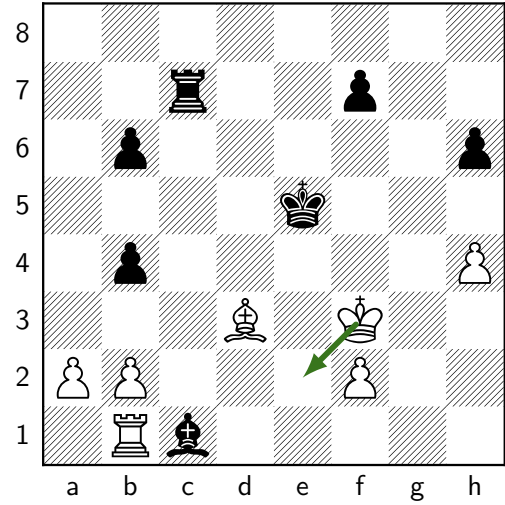
(a)



(b)

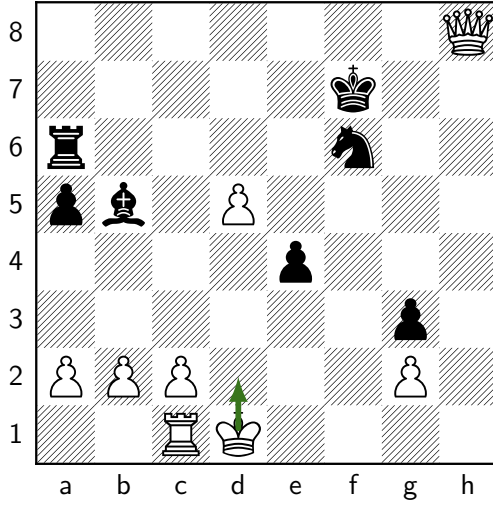


(c)

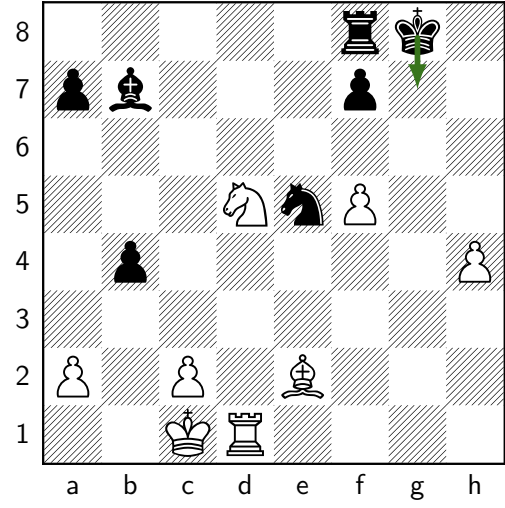


(d)

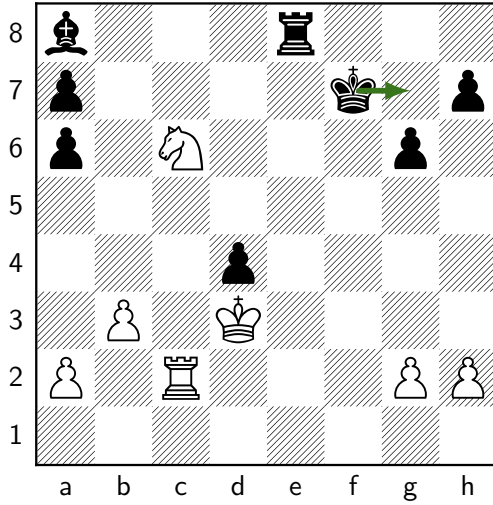
Figure 28: **Move 35, White: Kd1.** (a) Our position; Melanie plays Kd1 (d2d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2406` (ply 69) [structural distance $d=42$, QAP obj=39]: the master played Kg7; under μ our Kd2 plays the role of that game’s Kg1; cross-type roles: our Qh8 as their Bb7, our Pb2 as their Ne5; transferred move d2d1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1722` (ply 79) [structural distance $d=42$, QAP obj=37]: the master played Kg7; under μ our Kd2 plays the role of that game’s Kf2; cross-type roles: our Qh8 as their Ba8, our bb5 as their Pa6; transferred move d2d1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#445` (ply 66) [structural distance $d=44$, QAP obj=14]: the master played Ke2; under μ our Kd2 plays the role of that game’s Kf3; cross-type roles: our Qh8 as their Rb1, our nf6 as their bc1; transferred move d2d1 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kd1 (summed QAP objective 101, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



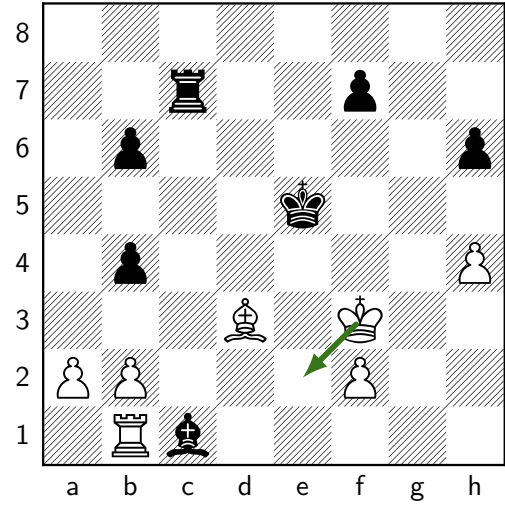
(a)



(b)

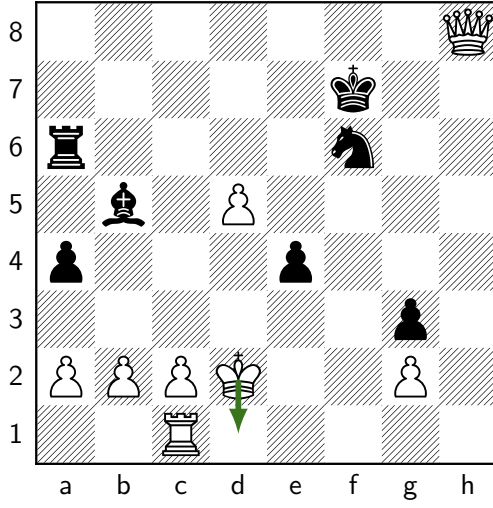


(c)

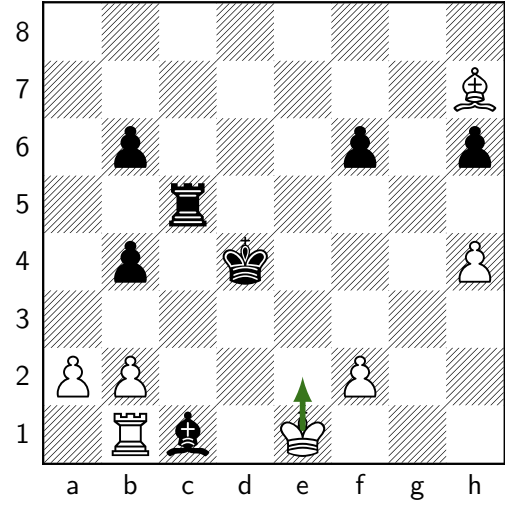


(d)

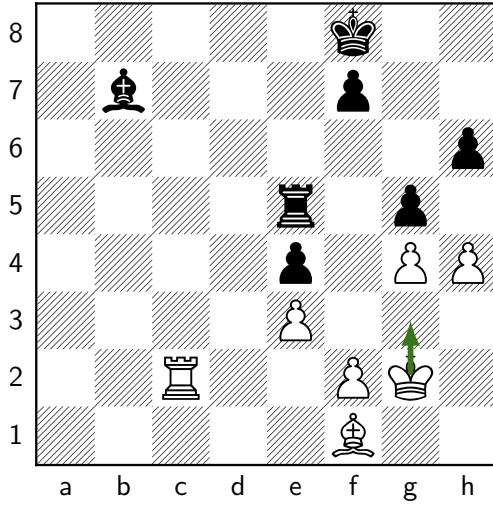
Figure 29: **Move 36, White: Kd2.** (a) Our position; Melanie plays Kd2 (d1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2406` (ply 69) [structural distance $d=38$, QAP obj=41]: the master played Kg7; under μ our Kd1 plays the role of that game’s Kg1; cross-type roles: our Qh8 as their Bb7, our Pg2 as their Ne5; transferred move d1d2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1722` (ply 79) [structural distance $d=43$, QAP obj=35]: the master played Kg7; under μ our Kd1 plays the role of that game’s Kf2; cross-type roles: our Qh8 as their Ba8, our bb5 as their Pa6; transferred move d1d2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#445` (ply 66) [structural distance $d=41$, QAP obj=16]: the master played Ke2; under μ our Kd1 plays the role of that game’s Kf3; cross-type roles: our Qh8 as their Rb1, our nf6 as their bc1; transferred move d1d2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kd2 (summed QAP objective 107, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



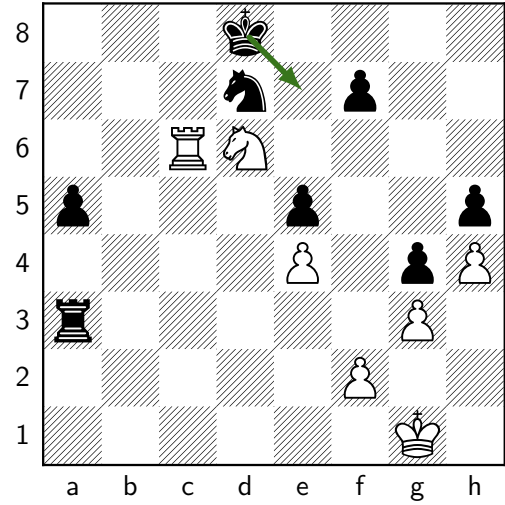
(a)



(b)

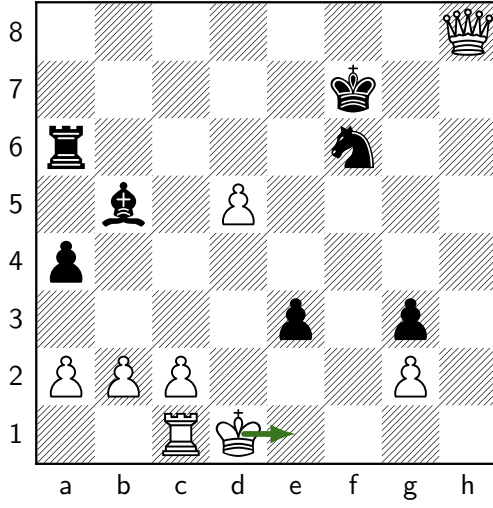


(c)

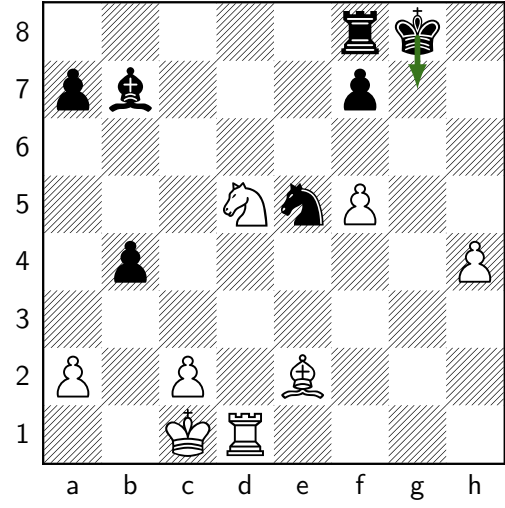


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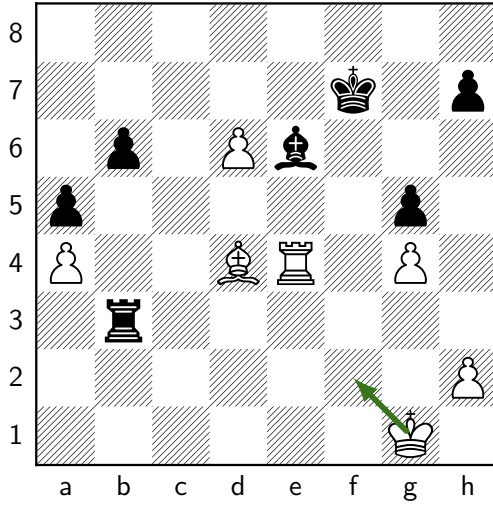
Figure 30: **Move 37, White: Kd1**. (a) Our position; Melanie plays Kd1 (d2d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#445` (ply 80) [structural distance $d=49$, QAP obj=18]: the master played Ke2; under μ our Kd2 plays the role of that game’s Ke1; cross-type roles: our Qh8 as their Rb1, our nf6 as their bc1; transferred move d2d1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#393` (ply 70) [structural distance $d=49$, QAP obj=14]: the master played Kg3; under μ our Kd2 plays the role of that game’s Kg2; cross-type roles: our Pb2 as their Bf1, our Pa2 as their Rc2; transferred move d2d1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#133` (ply 83) [structural distance $d=49$, QAP obj=10]: the master played Ke7; under μ our Kd2 plays the role of that game’s Kd1; cross-type roles: our Rc1 as their Nd7, our Qh8 as their Pe5; transferred move d2d1 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kd1 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



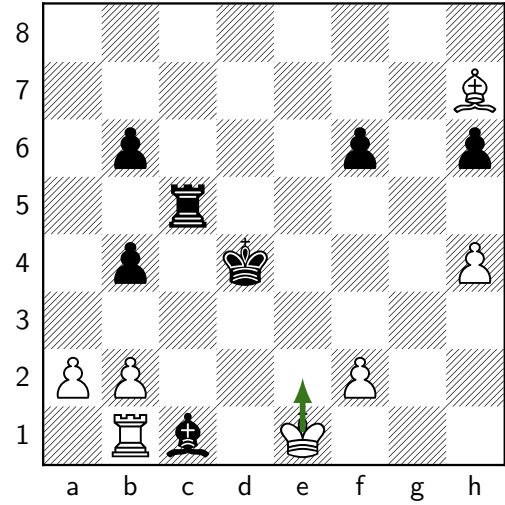
(a)



(b)

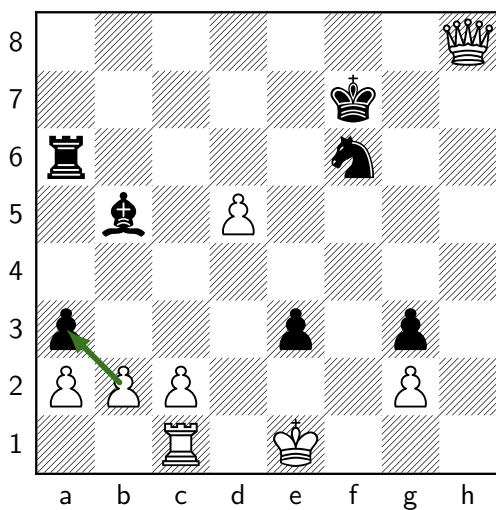


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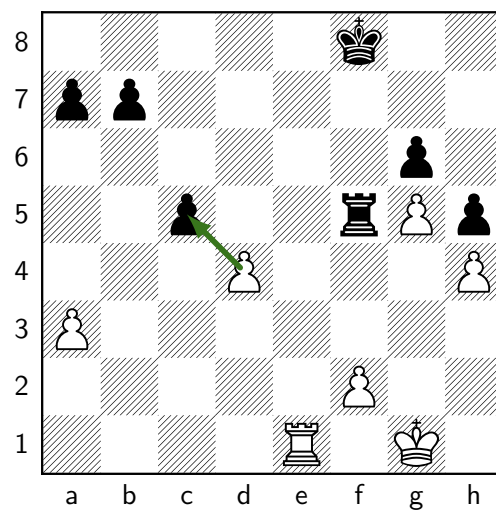


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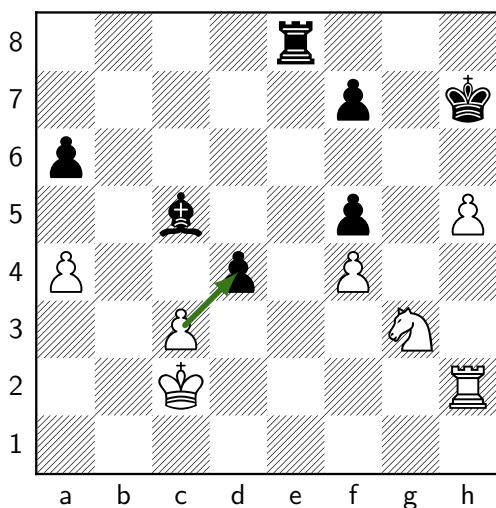
Figure 31: **Move 38, White: Ke1.** (a) Our position; Melanie plays Ke1 (d1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2406` (ply 69) [structural distance $d=39$, QAP obj=40]: the master played Kg7; under μ our Kd1 plays the role of that game’s Kg1; cross-type roles: our Qh8 as their Bb7, our Pb2 as their Ne5; transferred move d1e1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2124` (ply 76) [structural distance $d=44$, QAP obj=35]: the master played Kf2; under μ our Kd1 plays the role of that game’s Kg1; cross-type roles: our Pc2 as their rb3, our Rc1 as their Bd4; transferred move d1e1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#445` (ply 80) [structural distance $d=46$, QAP obj=16]: the master played Ke2; under μ our Kd1 plays the role of that game’s Ke1; cross-type roles: our Qh8 as their Rb1, our nf6 as their bc1; transferred move d1e1 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Ke1 (summed QAP objective 183, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



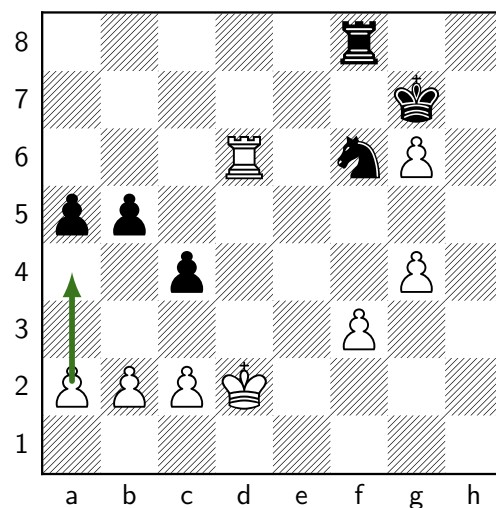
(a)



(b)

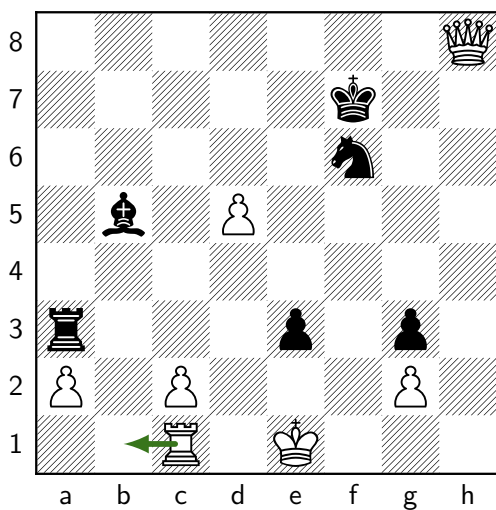


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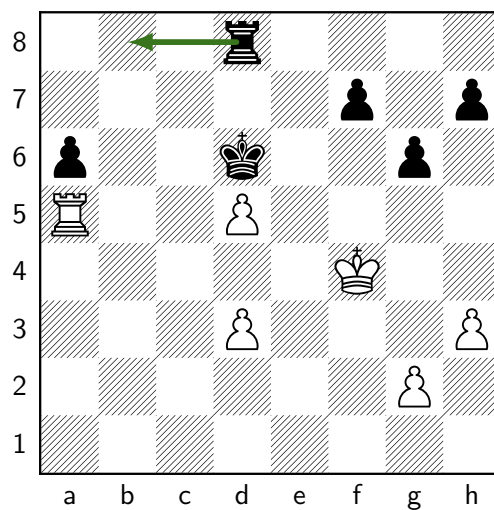


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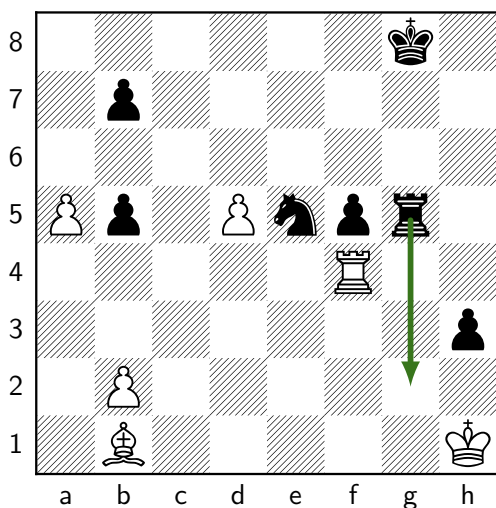
Figure 32: **Move 39, White: bxa3**. (a) Our position; Melanie plays **bxa3** (**b2a3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2315` (ply 60) [structural distance $d=50$, QAP obj=39]: the master played **dxc5**; under μ our *Pb2* plays the role of that game’s *Pd4*; cross-type roles: our *ra6* as their *Ph4*, our *Qh8* as their *rf5*; transferred move **b2a3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1113` (ply 82) [structural distance $d=49$, QAP obj=33]: the master played **cxd4**; under μ our *Pb2* plays the role of that game’s *Pc3*; cross-type roles: our *Qh8* as their *Ng3*, our *ra6* as their *bc5*; transferred move **b2a3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1303` (ply 76) [structural distance $d=39$, QAP obj=32]: the master played **a4**; under μ our *Pb2* plays the role of that game’s *Pa2*; cross-type roles: our *Rc1* as their *pa5*, our *Qh8* as their *Rd6*; transferred move **b2a3** (**endorses this move**). Synthesis: 5 retrieved analogs transferred to **bxa3** (summed QAP objective 157, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



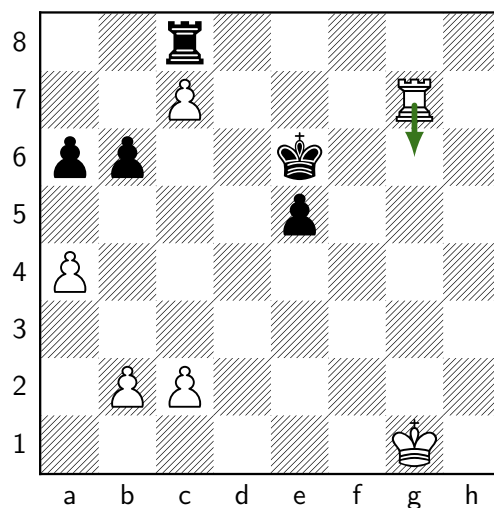
(a)



(b)

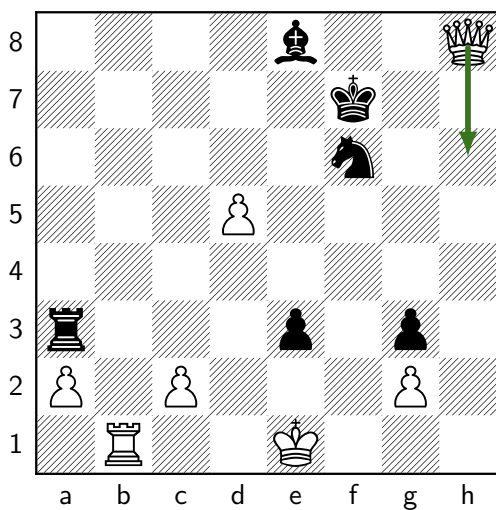


(c)

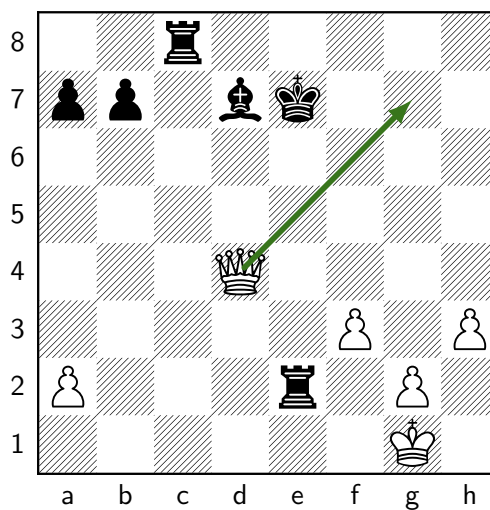


(d)

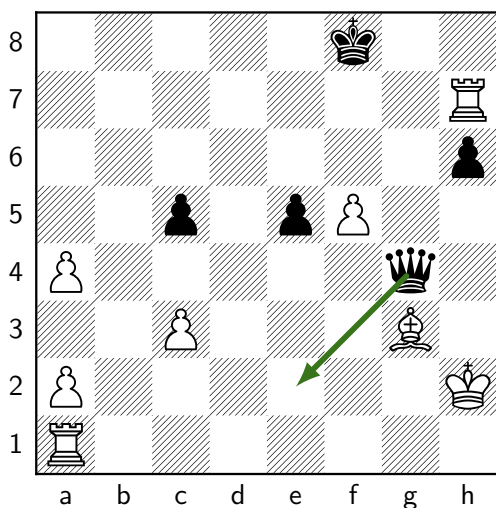
Figure 33: **Move 40, White: Rb1.** (a) Our position; Melanie plays Rb1 (c1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2358` (ply 69) [structural distance $d=35$, QAP obj=17]: the master played Rb8; under μ our *Rc1* plays the role of that game’s *Rd1*; cross-type roles: our *nf6* as their *Ph7*, our *Qh8* as their *Pg6*; transferred move c1b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3721` (ply 77) [structural distance $d=29$, QAP obj=14]: the master played Rg2; under μ our *Rc1* plays the role of that game’s *Rg4*; cross-type roles: our *Qh8* as their *Pb7*, our *nf6* as their *pa5*; transferred move c1b1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3578` (ply 72) [structural distance $d=34$, QAP obj=13]: the master played Rg6+; under μ our *Rc1* plays the role of that game’s *Rg7*; cross-type roles: our *nf6* as their *Pa4*, our *Qh8* as their *pa6*; transferred move c1b1 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Rb1 (summed QAP objective 90, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



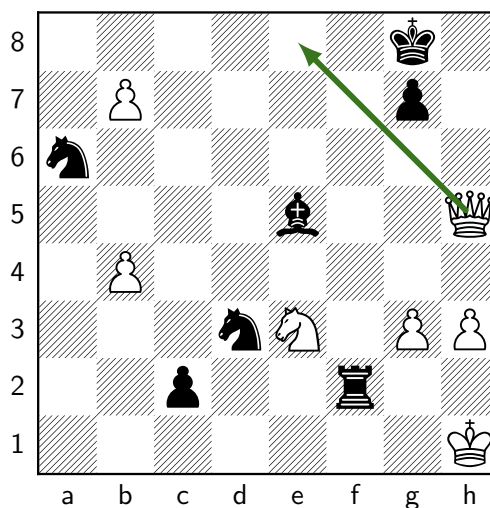
(a)



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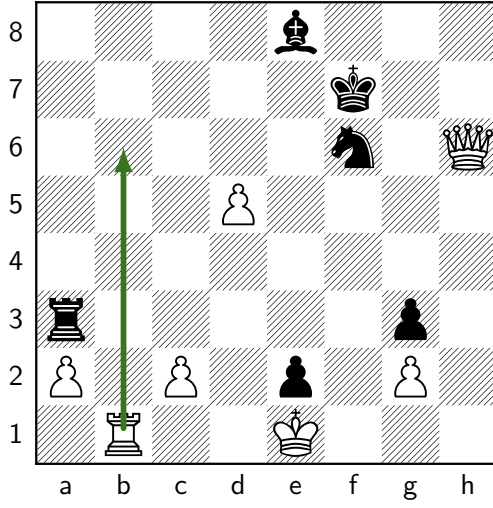


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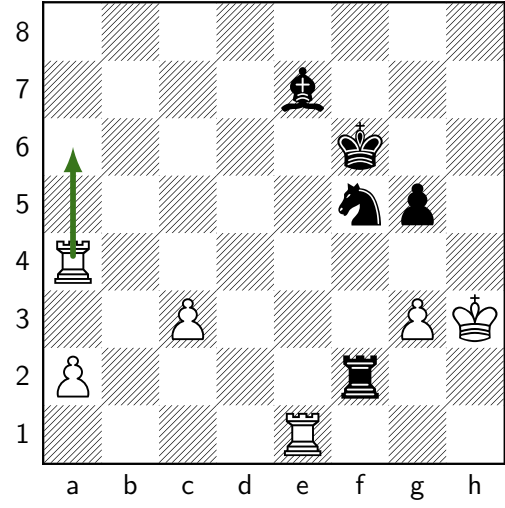


(d)

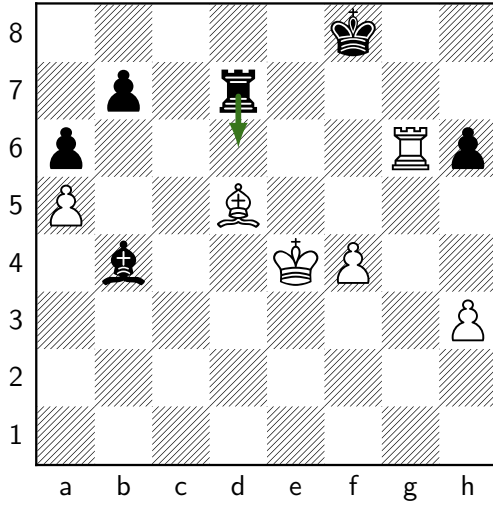
Figure 34: **Move 41, White: Qh6.** (a) Our position; Melanie plays Qh6 (h8h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#921` (ply 70) [structural distance $d=47$, QAP obj=43]: the master played Qg7+; under μ our Qh8 plays the role of that game’s Qd4; cross-type roles: our Rb1 as their Pa2, our nf6 as their pa7; transferred move h8h6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#718` (ply 87) [structural distance $d=37$, QAP obj=37]: the master played Qe2+; under μ our Qh8 plays the role of that game’s Qg5; cross-type roles: our be8 as their pf5, our nf6 as their bg3; transferred move h8h6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#709` (ply 96) [structural distance $d=45$, QAP obj=36]: the master played Qe8+; under μ our Qh8 plays the role of that game’s Qh5; cross-type roles: our pe3 as their rf2, our Qh8 as their nd3; transferred move h8h6 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Qh6 (summed QAP objective 228, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



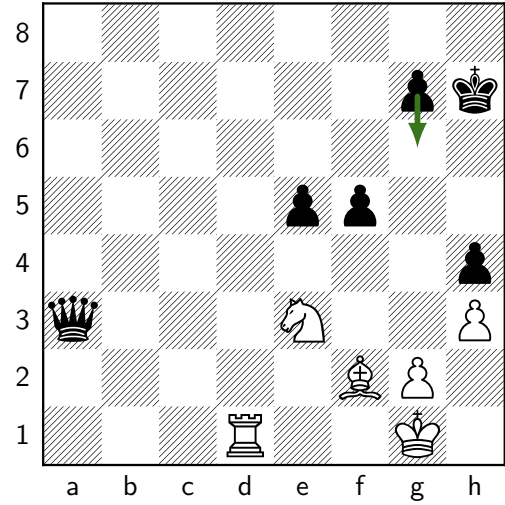
(a)



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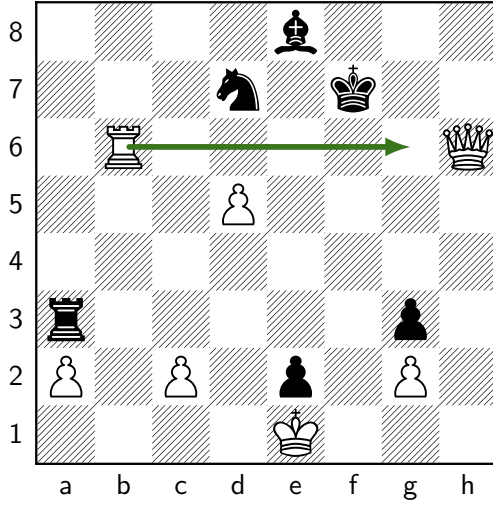


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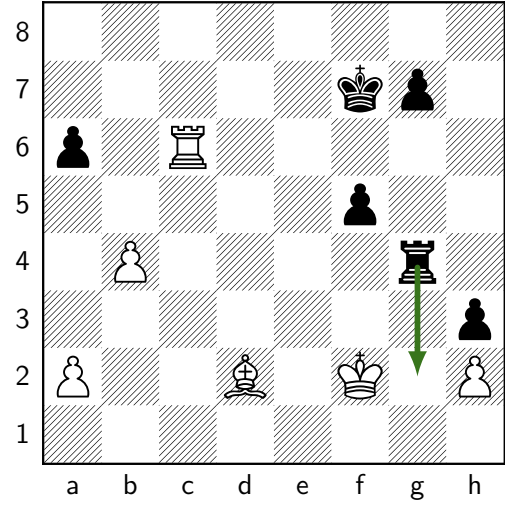


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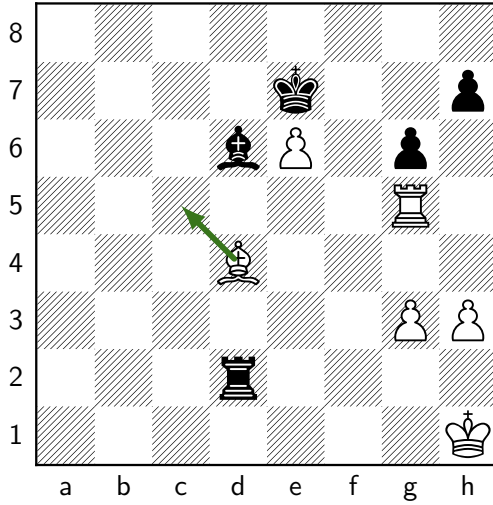
Figure 35: **Move 42, White: Rb6.** (a) Our position; Melanie plays Rb6 (b1b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#224` (ply 66) [structural distance $d=47$, QAP obj=14]: the master played Ra6+; under μ our Rb1 plays the role of that game’s Ra4; cross-type roles: our Qh6 as their Re1, our Pc2 as their rf2; transferred move b1b6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2916` (ply 73) [structural distance $d=44$, QAP obj=53]: the master played Rd6; under μ our Qh6 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our be8 as their Pa6, our Ke1 as their rg6; transferred move h6e3. (d) Retrieved from `ccrl_top_p0002#3570` (ply 85) [structural distance $d=47$, QAP obj=37]: the master played g6; under μ our Rb1 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our ra3 as their bf2, our be8 as their rd1; transferred move b1d1. Synthesis: 1 retrieved analog transferred to Rb6 (summed QAP objective 14, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



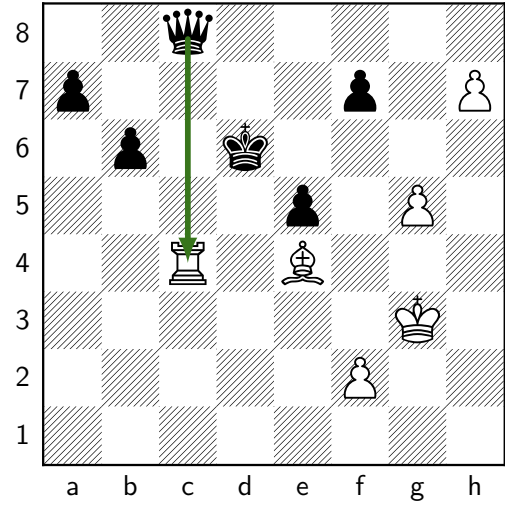
(a)



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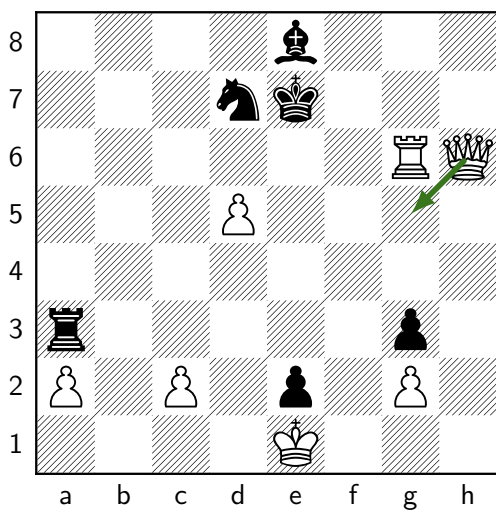


(c)

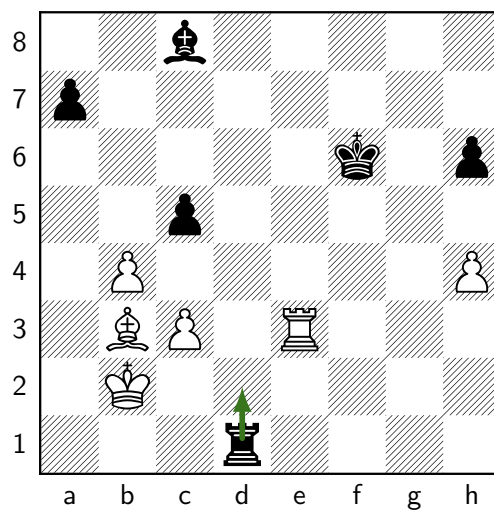


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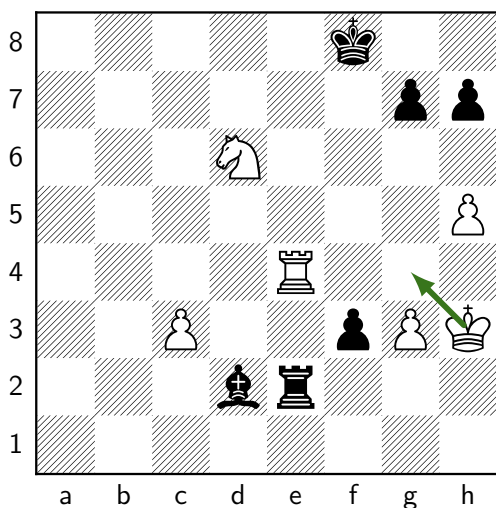
Figure 36: **Move 43, White: Rg6.** (a) Our position; Melanie plays Rg6 (b6g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2448` (ply 77) [structural distance $d=49$, QAP obj=3]: the master played Rg2+; under μ our Rb6 plays the role of that game’s Rg5; cross-type roles: our ra3 as their Pa6, our pe2 as their rc6; transferred move b6g6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#765` (ply 78) [structural distance $d=50$, QAP obj=21]: the master played Bc5; under μ our Rb6 plays the role of that game’s Bd4 (a **cross-type** correspondence); cross-type roles: our nd7 as their rd2, our Qh6 as their Pe6; transferred move b6d6. (d) Retrieved from `ccrl_top_p0002#3799` (ply 97) [structural distance $d=46$, QAP obj=18]: the master played Qxc4; under μ our Rb6 plays the role of that game’s Qc1 (a **cross-type** correspondence); cross-type roles: our ra3 as their Pb6, our nd7 as their rc4; transferred move b6b5. Synthesis: 1 retrieved analog transferred to Rg6 (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



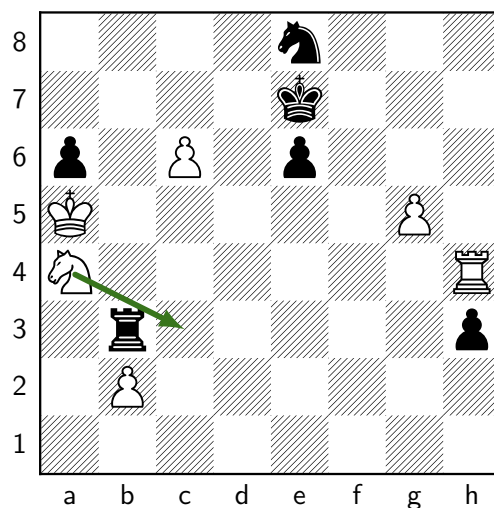
(a)



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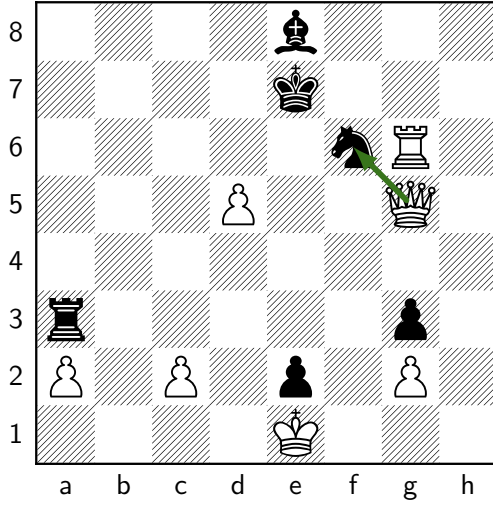


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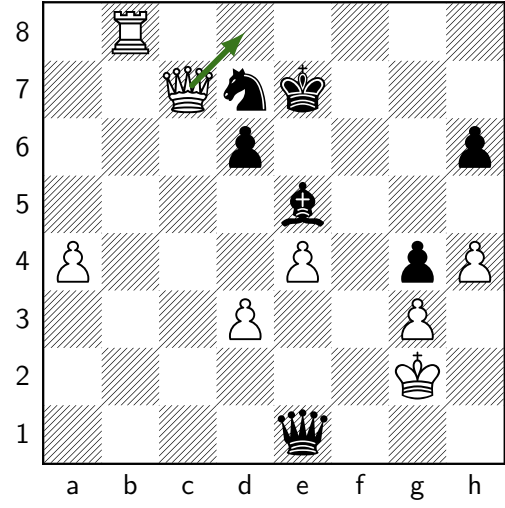


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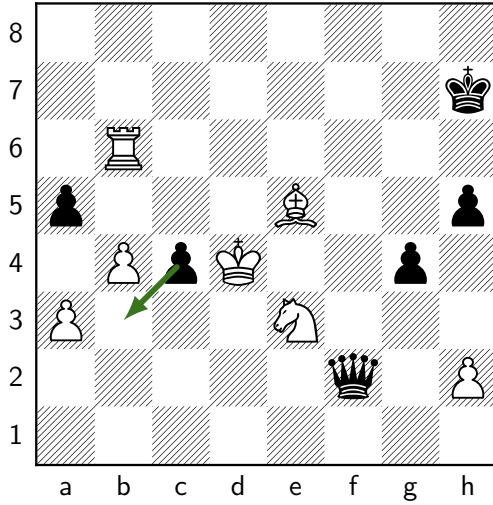
Figure 37: **Move 44, White: Qg5+.** (a) Our position; Melanie plays Qg5+ (h6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3510` (ply 85) [structural distance $d=50$, QAP obj=30]: the master played Rd2+; under μ our Rg6 plays the role of that game’s Rd8; cross-type roles: our Pd5 as their Bc8, our Qh6 as their Pa7; transferred move g6g8. (c) Retrieved from `ccr1_top_p0002#2870` (ply 78) [structural distance $d=51$, QAP obj=28]: the master played Kg4; under μ our Ke1 plays the role of that game’s Kh3; cross-type roles: our ra3 as their bd2, our be8 as their re2; transferred move e1d2. (d) Retrieved from `ccr1_top_p0003#1657` (ply 94) [structural distance $d=45$, QAP obj=24]: the master played Nc3; under μ our Pc2 plays the role of that game’s Na4 (a **cross-type** correspondence); cross-type roles: our Rg6 as their ph3, our be8 as their Rh4; transferred move c2c3. Synthesis: no retrieved analog transferred to Qg5+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



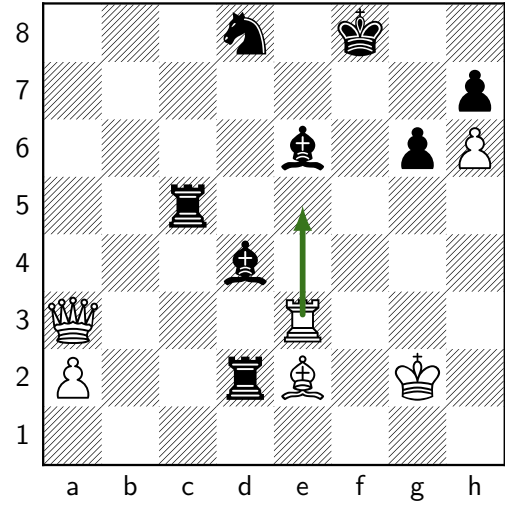
(a)



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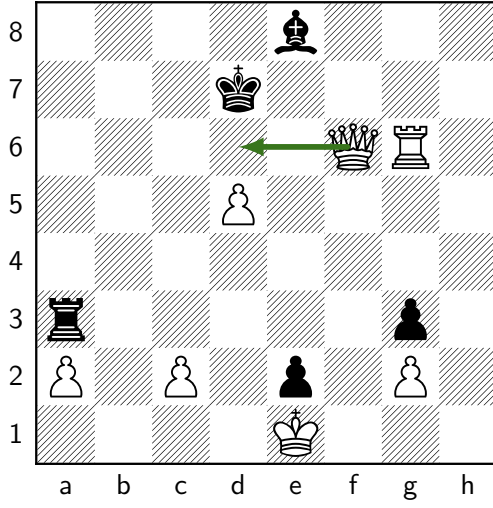


(c)

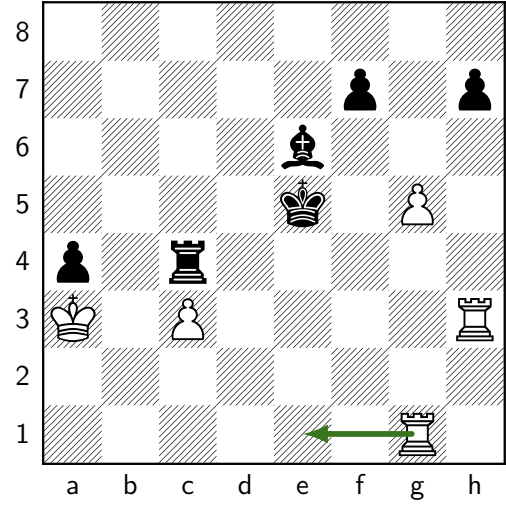


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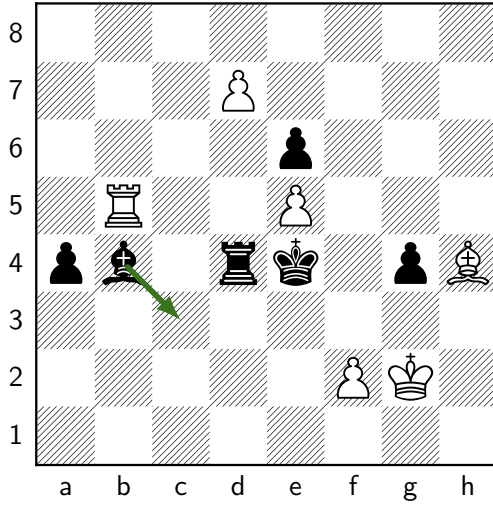
Figure 38: **Move 45, White: Qxf6+.** (a) Our position; Melanie plays Qxf6+ (g5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#557` (ply 90) [structural distance $d=65$, QAP obj=58]: the master played Qd8+; under μ our Qg5 plays the role of that game’s Qc7; cross-type roles: our ra3 as their qe1, our nf6 as their pd6; transferred move g5f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2093` (ply 83) [structural distance $d=61$, QAP obj=77]: the master played cxb3; under μ our Pd5 plays the role of that game’s Pc5; cross-type roles: our pe2 as their rb6, our ra3 as their Pa5; transferred move d5d6. (d) Retrieved from `ccr1_top_p0003#2846` (ply 78) [structural distance $d=62$, QAP obj=76]: the master played Re5; under μ our Rg6 plays the role of that game’s Re3; cross-type roles: our Qg5 as their rd2, our nf6 as their Be2; transferred move g6g8. Synthesis: 1 retrieved analog transferred to Qxf6+ (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



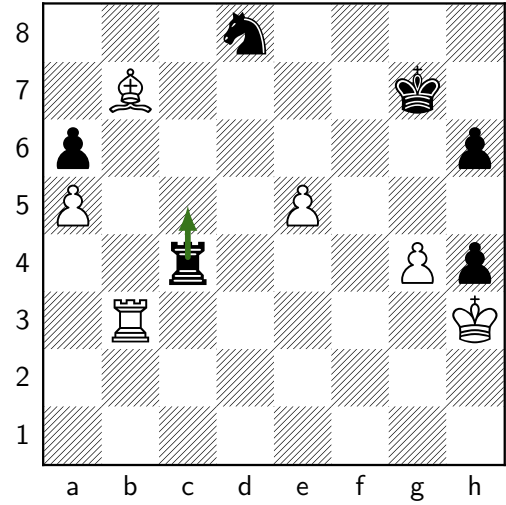
(a)



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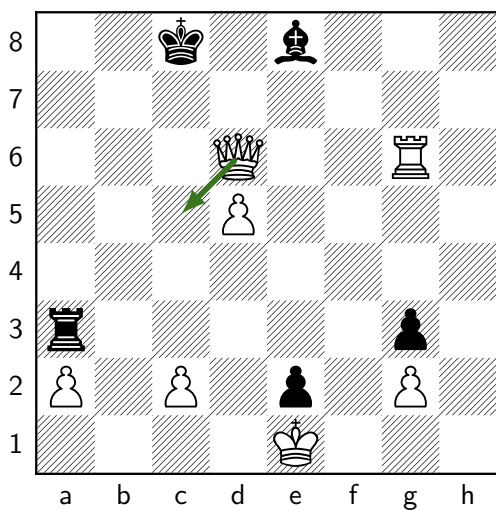


(c)

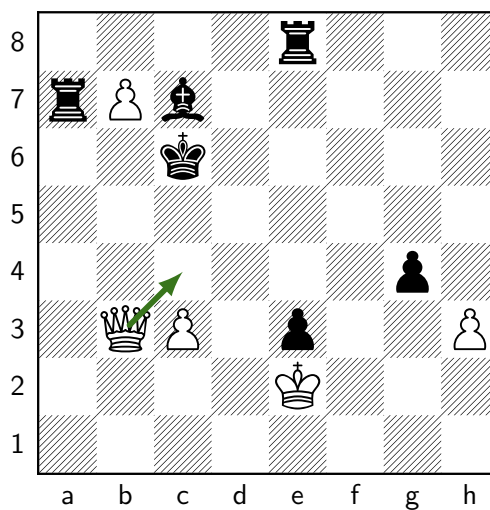


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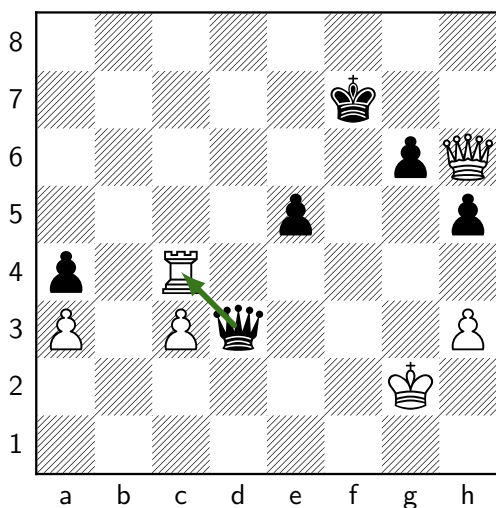
Figure 39: **Move 46, White: Qd6+.** (a) Our position; Melanie plays Qd6+ (f6d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1279` (ply 94) [structural distance $d=44$, QAP obj=39]: the master played Re1+; under μ our *Pc2* plays the role of that game’s *Rg1* (a **cross-type** correspondence); cross-type roles: our *Qf6* as their *pf7*; transferred move c2c4. (c) Retrieved from `ccrl_top_p0002#1081` (ply 107) [structural distance $d=44$, QAP obj=22]: the master played Bc3; under μ our *Rg6* plays the role of that game’s *Bb5* (a **cross-type** correspondence); cross-type roles: our *be8* as their *rb5*, our *Qf6* as their *Rd4*; transferred move g6h6. (d) Retrieved from `ccrl_top_p0003#699` (ply 95) [structural distance $d=40$, QAP obj=22]: the master played Rc5; under μ our *Pc2* plays the role of that game’s *Rc5* (a **cross-type** correspondence); cross-type roles: our *be8* as their *Nd8*, our *Rg6* as their *bb7*; transferred move c2c4. Synthesis: no retrieved analog transferred to Qd6+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



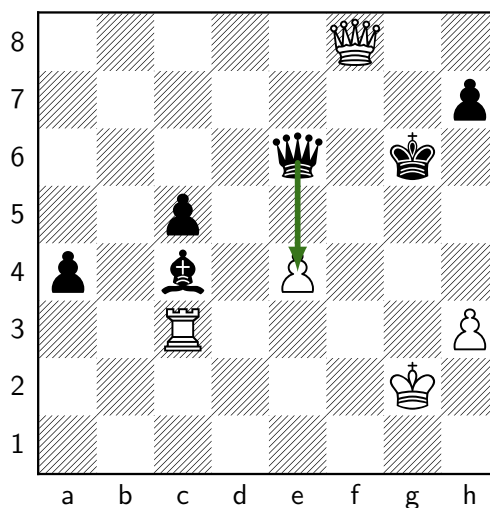
(a)



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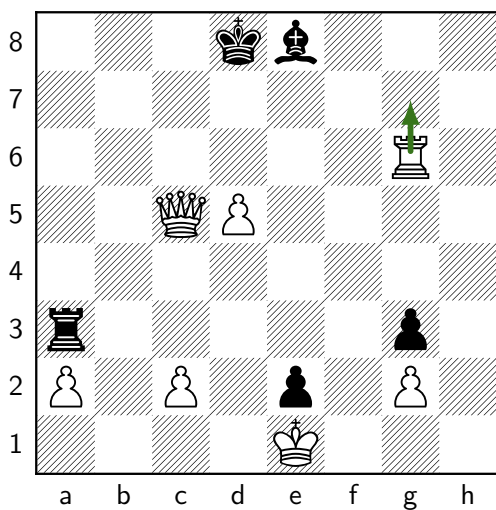


(c)

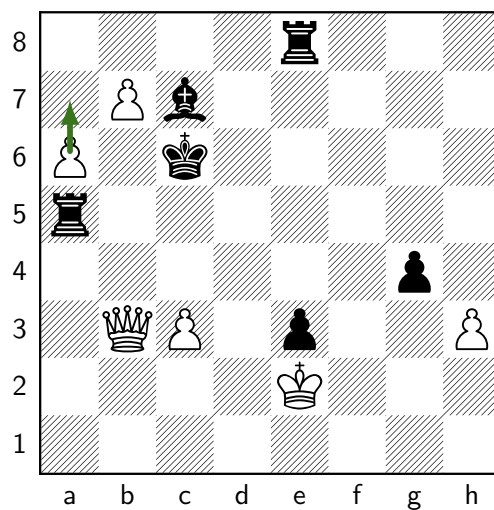


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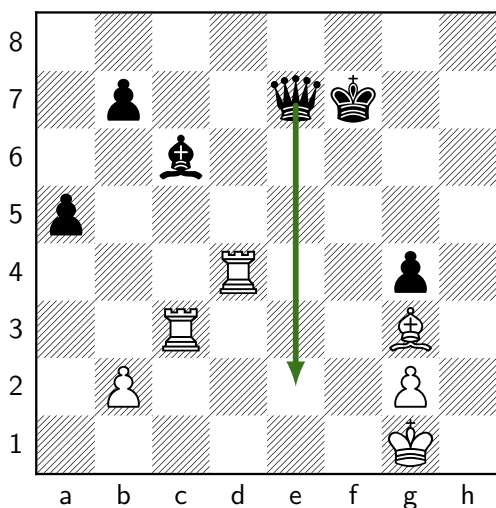
Figure 40: **Move 47, White: Qc5+.** (a) Our position; Melanie plays Qc5+ (d6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1053` (ply 108) [structural distance $d=42$, QAP obj=24]: the master played Qc4+; under μ our Qd6 plays the role of that game's Qb3; cross-type roles: our ra3 as their pg4, our be8 as their ra7; transferred move d6c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2645` (ply 99) [structural distance $d=49$, QAP obj=21]: the master played Qxc4; under μ our Qd6 plays the role of that game's Qd6; cross-type roles: our Rg6 as their Pg6, our pe2 as their qh6; transferred move d6c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1477` (ply 101) [structural distance $d=51$, QAP obj=19]: the master played Qxe4+; under μ our Qd6 plays the role of that game's Qe3; cross-type roles: our pe2 as their qf8, our Rg6 as their Bc4; transferred move d6c5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qc5+ (summed QAP objective 64, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



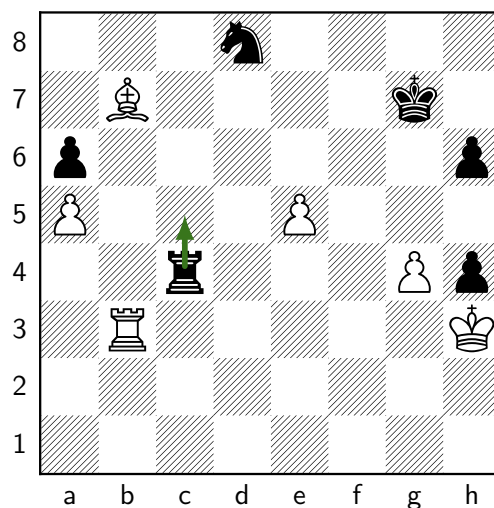
(a)



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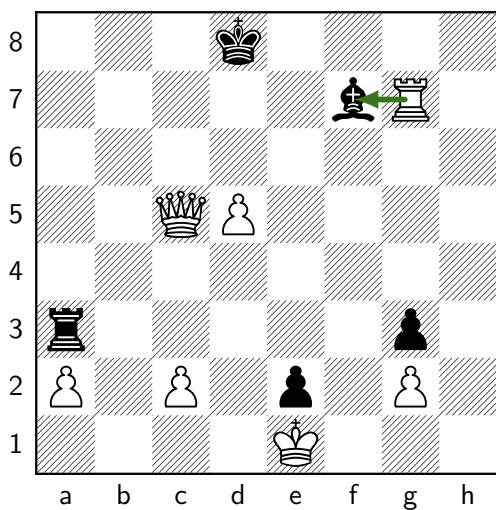


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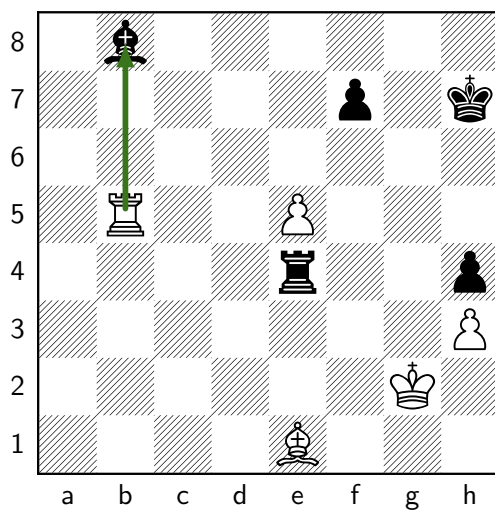


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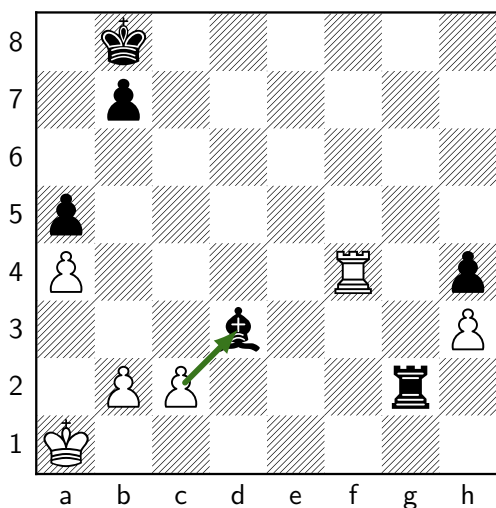
Figure 41: **Move 48, White: Rg7.** (a) Our position; Melanie plays Rg7 (g6g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#1053 (ply 106) [structural distance $d=24$, QAP obj=24]: the master played a7; under μ our Rg6 plays the role of that game’s Pa6 (a **cross-type** correspondence); cross-type roles: our ra3 as their pg4, our be8 as their ra5; transferred move g6g7 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#2301 (ply 101) [structural distance $d=40$, QAP obj=22]: the master played Qe2; under μ our Rg6 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our ra3 as their Bc6, our be8 as their rd4; transferred move g6g7 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#699 (ply 95) [structural distance $d=41$, QAP obj=35]: the master played Rc5; under μ our Pd5 plays the role of that game’s Rc5 (a **cross-type** correspondence); cross-type roles: our Qc5 as their Nd8, our ra3 as their bb7; transferred move d5d6. Synthesis: 2 retrieved analogs transferred to Rg7 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



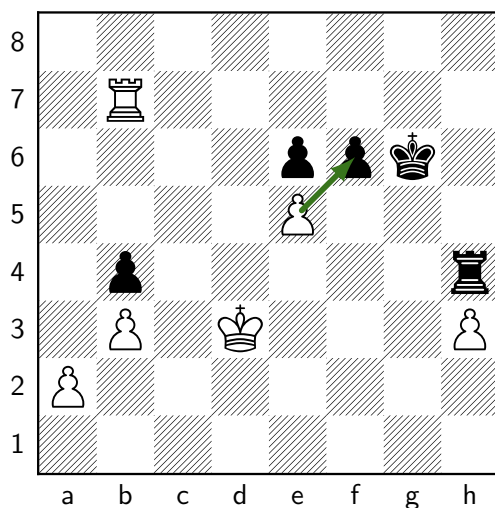
(a)



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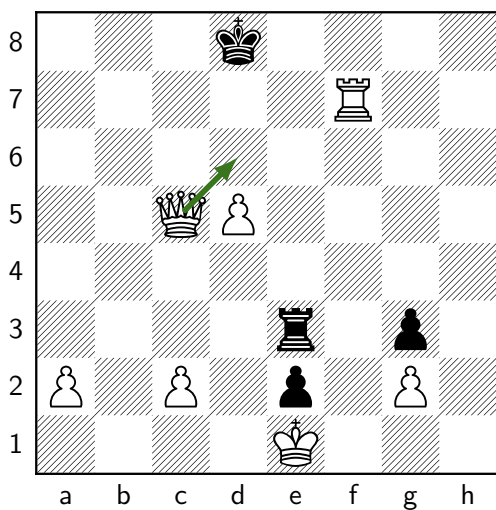


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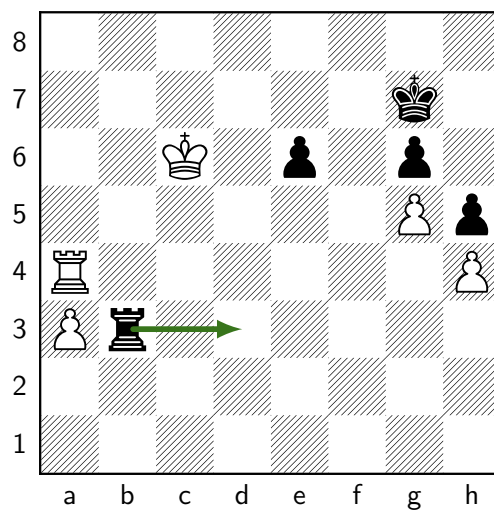


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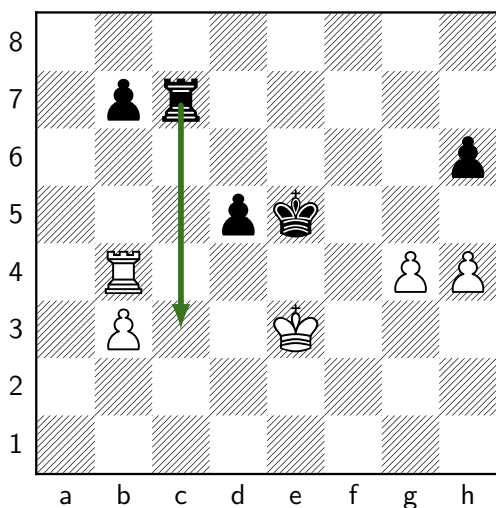
Figure 42: **Move 49, White: Rxf7**. (a) Our position; Melanie plays Rxf7 (g7f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3886` (ply 90) [structural distance $d=39$, QAP obj=45]: the master played Rxb8; under μ our Rg7 plays the role of that game’s Rb5; cross-type roles: our ra3 as their Be1, our Qc5 as their re4; transferred move g7f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#845` (ply 82) [structural distance $d=41$, QAP obj=19]: the master played cxd3; under μ our Rg7 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Qc5 as their Rf4; transferred move g7f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1299` (ply 84) [structural distance $d=39$, QAP obj=14]: the master played exf6; under μ our Rg7 plays the role of that game’s Pe5 (a **cross-type** correspondence); cross-type roles: our Qc5 as their Pa2, our bf7 as their pf6; transferred move g7f7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rxf7 (summed QAP objective 95, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



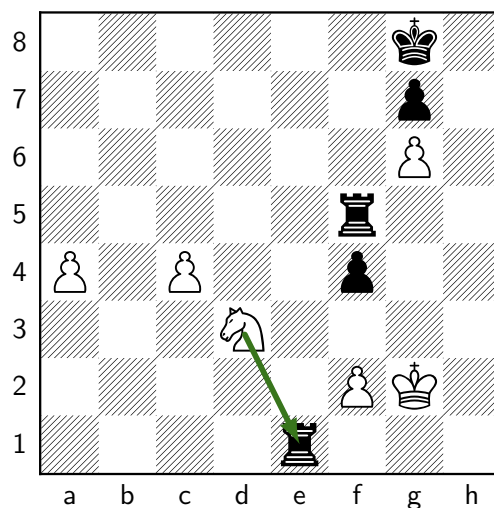
(a)



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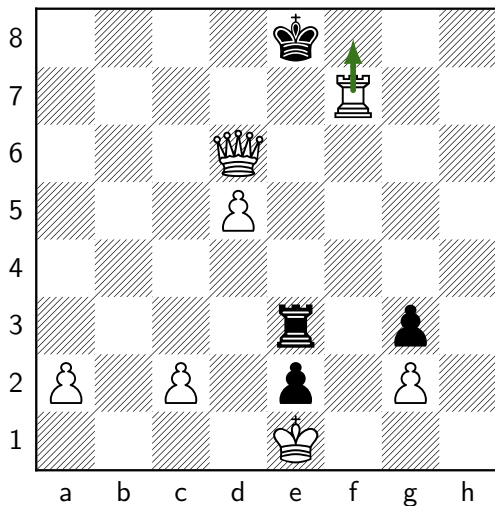


(c)

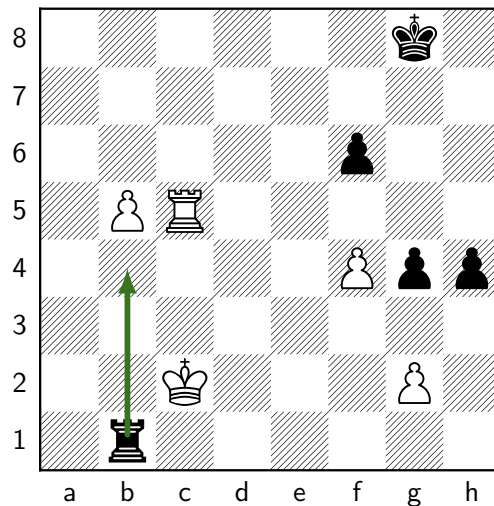


(d)

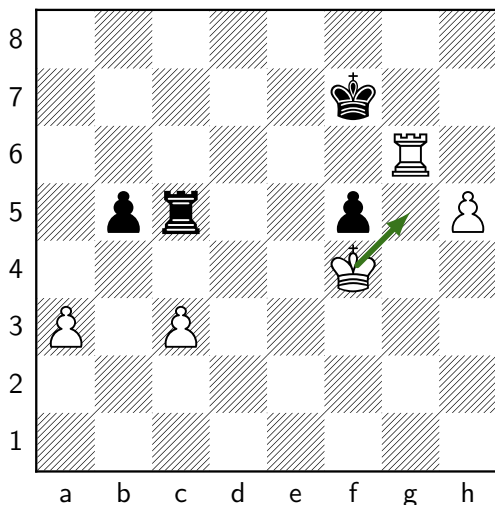
Figure 43: **Move 50, White: Qd6+.** (a) Our position; Melanie plays Qd6+ (c5d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1601` (ply 91) [structural distance $d=36$, QAP obj=9]: the master played Rd3; under μ our Qc5 plays the role of that game's Rb6 (a **cross-type** correspondence); cross-type roles: our re3 as their pa3; transferred move c5d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2971` (ply 111) [structural distance $d=31$, QAP obj=3]: the master played Rc3+; under μ our Qc5 plays the role of that game's Rc2 (a **cross-type** correspondence); transferred move c5d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#973` (ply 94) [structural distance $d=35$, QAP obj=21]: the master played Nxex1; under μ our Qc5 plays the role of that game's Nd3 (a **cross-type** correspondence); cross-type roles: our pg3 as their rf5, our Rf7 as their Pg6; transferred move c5b4. Synthesis: 2 retrieved analogs transferred to Qd6+ (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



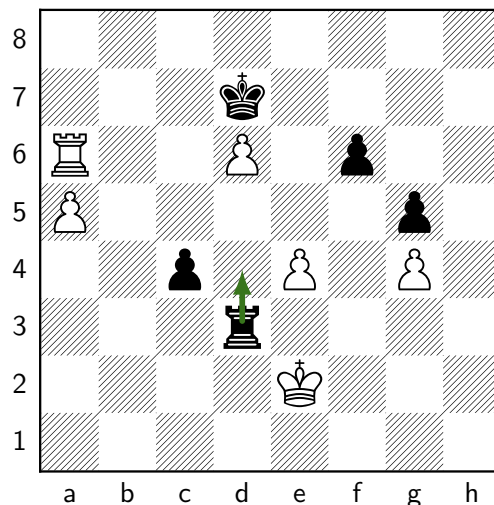
(a)



(b)



(c)



(d)

Figure 44: **Move 51, White: Rf8#.** (a) Our position; Melanie plays Rf8# (f7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0000#1811` (ply 105) [structural distance $d=36$, QAP obj=22]: the master played Rb4; under μ our Rf7 plays the role of that game’s Rb8; transferred move f7g7. (c) Retrieved from `ccr1_top.p0002#1988` (ply 100) [structural distance $d=37$, QAP obj=22]: the master played Kg5; under μ our Ke1 plays the role of that game’s Kf4; cross-type roles: our Qd6 as their Pa3; transferred move e1d2. (d) Retrieved from `ccr1_top.p0000#3112` (ply 87) [structural distance $d=39$, QAP obj=21]: the master played Rd4; under μ our Rf7 plays the role of that game’s Rd6; cross-type roles: our Qd6 as their Pg5; transferred move f7f5. Synthesis: no retrieved analog transferred to Rf8# – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.